

KKB 4.22MW - PVS Solaris Insight Billing & Monitoring System

WEB DESIGN SPECIFICATION

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						PROJECT NO.: TBC	
0	For Approval	Nattaphum	San	Chinnaphat	23-Jan-2026	PO.No.: PO-202601002	REV.
						Doc.No.: KKB-SAT-EN-01	
REV.	DESCRIPTION	BY	CHK.	APPR.	DATE	FM-OP01-09 (R00)	0



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Contents

PROPRIETARY NOTICE	2
1 INTRODUCTION.....	5
1.1 Purpose.....	5
2 Central Monitoring Web Design Overview	6
2.1 Navigation Panel	6
2.2 Overview.....	7
2.3 Performance	8
2.4 Trends.....	9
2.5 Report.....	10
2.6 Billing Admin.....	10
2.7 Billings	14
2.8 User Setting	14
2.9 Alarm Setting	20
2.10 Notifications Setting	22
2.11 Report Admin	25
3 Site Monitoring Web Design	26
3.1 Site Overview.....	26
3.2 Dashboard	27
3.3 Performance	28
3.4 Real-time	29
3.5 Diagram	29
3.6 Events	30
3.7 Charts	31
3.8 Reports	32
4 Billing System	33
4.1 Workflow Diagram.....	33
4.2 Billing Dashboard.....	34
4.3 Energy Confirmation (Step 1)	35
4.4 Invoice Management (Step 2).....	40
4.5 Signed Invoice Submission.....	42
4.6 Payment Submission (Step 3)	44
4.7 Payment Verification	46



4.8	Receipt Issuance (Step 4)	47
4.9	Final Billing Confirmation.....	48
4.10	Status & Monitoring	50
4.11	Billing Management (Admin Control)	51
4.12	Notification & Email Communication	53
4.13	Example Billing Design	56



1 INTRODUCTION

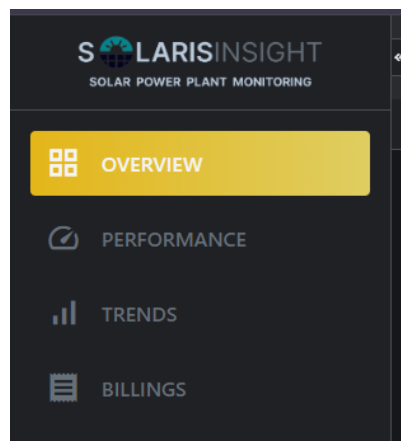
1.1 Purpose

The purpose of this Web Design Specification (WDS) is to define the designs principle and functionality of Web Monitoring and Billing to meet the design philosophy. Those items will be specifically identified.

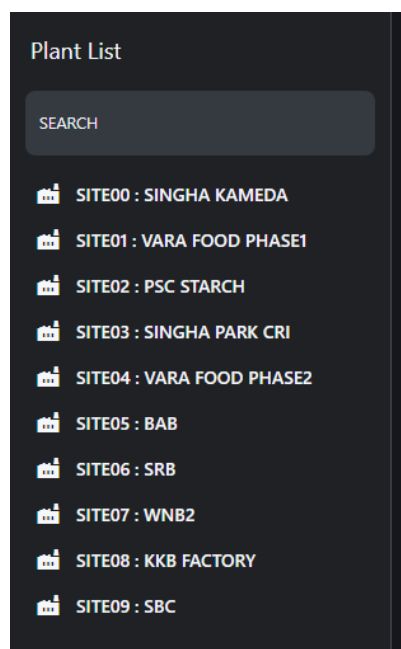
2 Central Monitoring Web Design Overview

2.1 Navigation Panel

Top Left Displays the *Solaris Insight* logo and the main navigation menu, consisting of: **Overview**, **Performance**, **Trends**, and **Billing**.

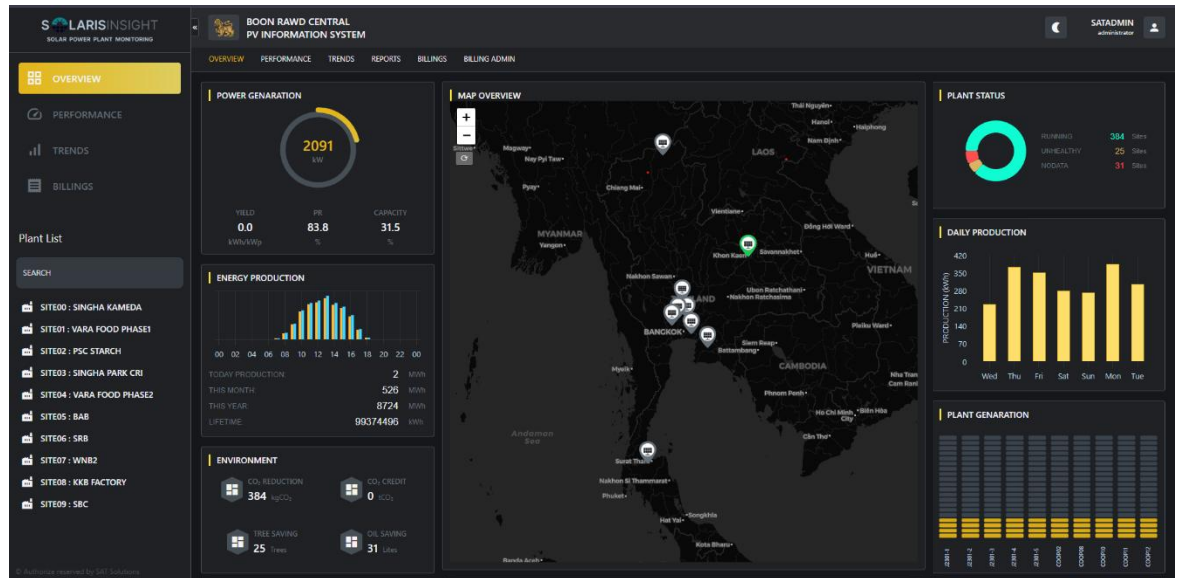


Left Panel Shows the list of sites available in the monitoring system. Users can click any site to view its corresponding details.

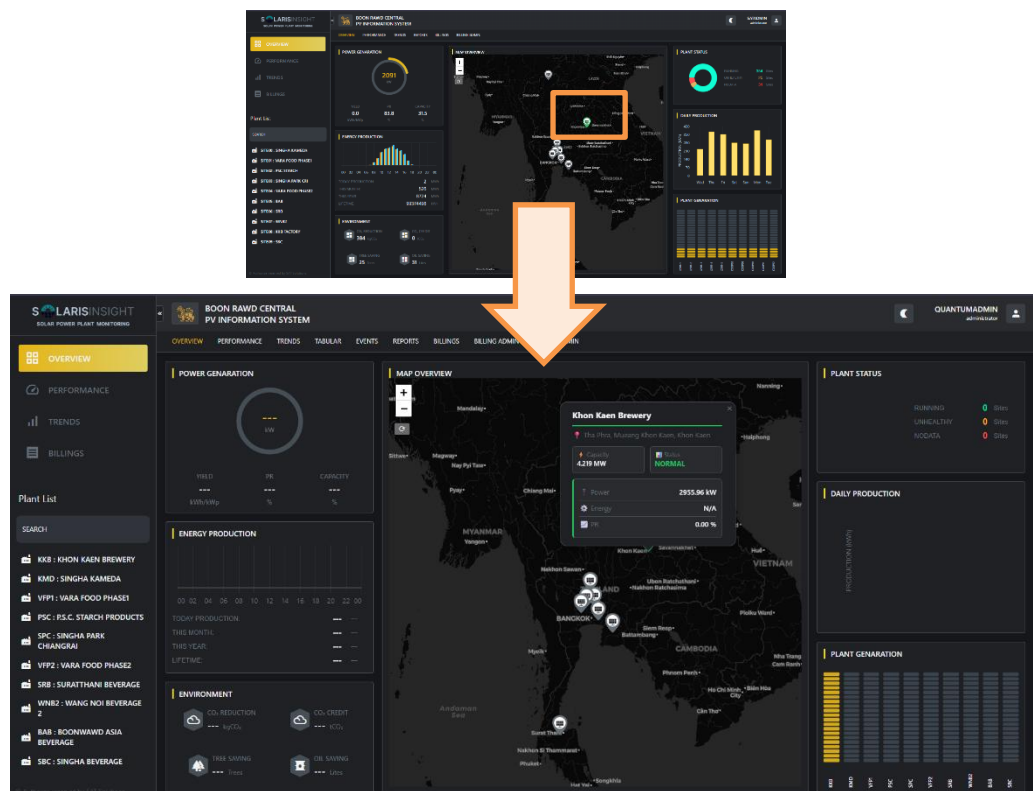


2.2 Overview

The Central Overview serves as a national-level dashboard for real-time monitoring of energy production across all sites. It presents system-wide performance indicators, including power output, energy production (daily, monthly, and cumulative), site status, and key performance metrics. A map view displays the location and performance of each region.

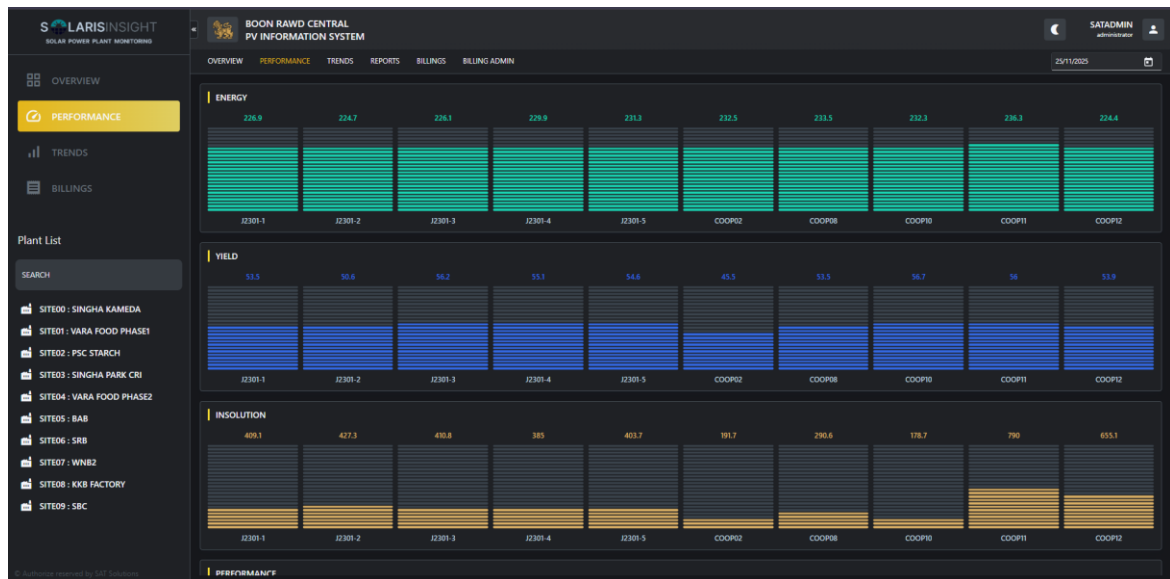


Map icons represent the location of each site. Users can click on an icon to view a summary of information and the current status of that site. The map also supports zooming in, zooming out, and resetting to the default view via controls located at the top-left corner.

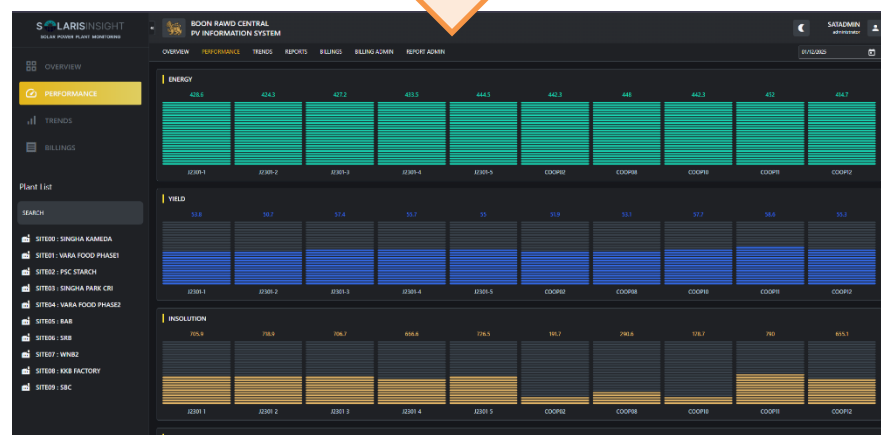
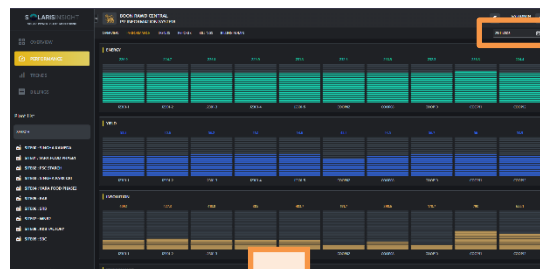


2.3 Performance

The Central Performance page displays comparative performance indicators such as energy production, yield, and other efficiency-related metrics for all sites.

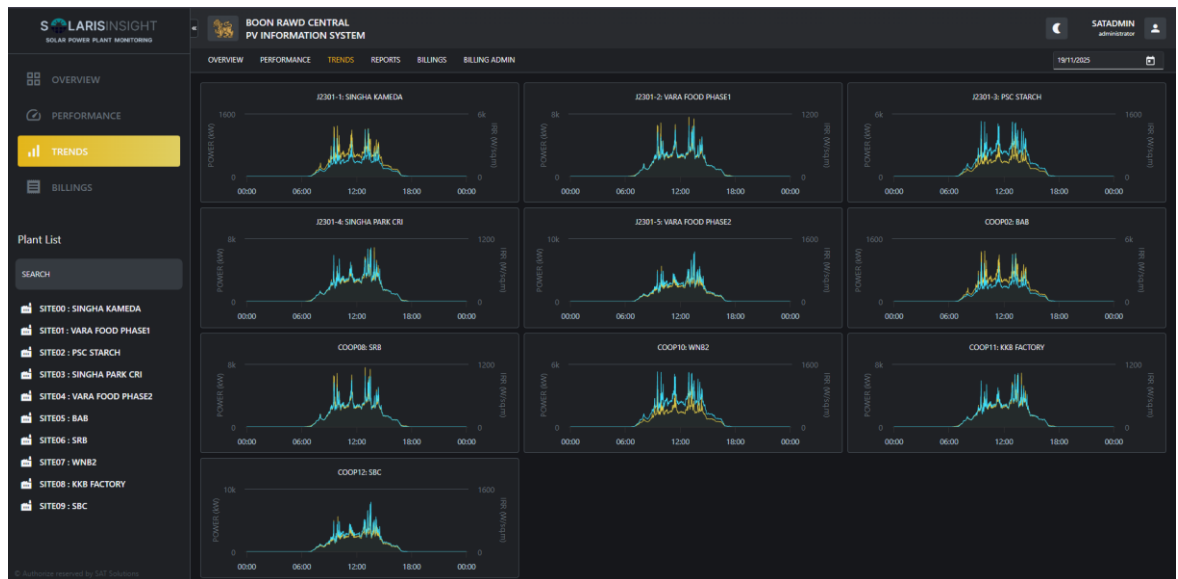


Users can review historical performance by selecting dates through the calendar button on the top-right corner.

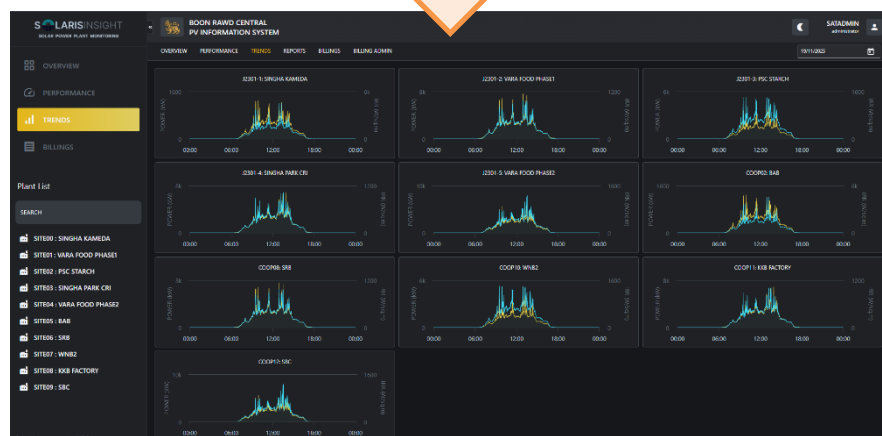


2.4 Trends

The Trends page displays comparative graphs of **Power Generation vs. Irradiance** across different sites to help identify abnormal production patterns using solar production profiles.

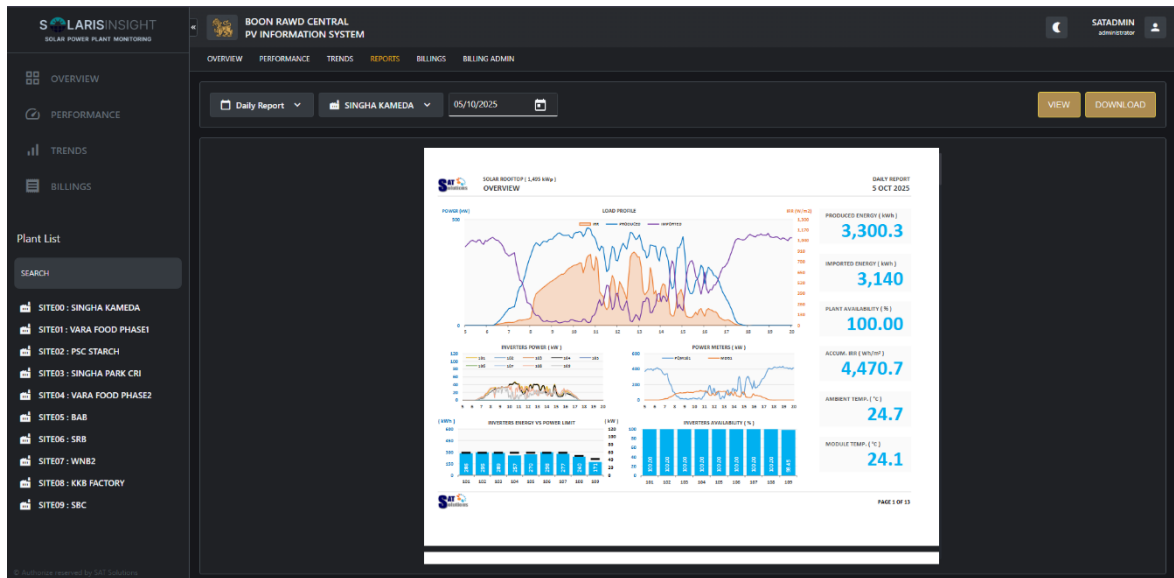


Users can review historical performance by selecting dates through the calendar button on the top-right corner.



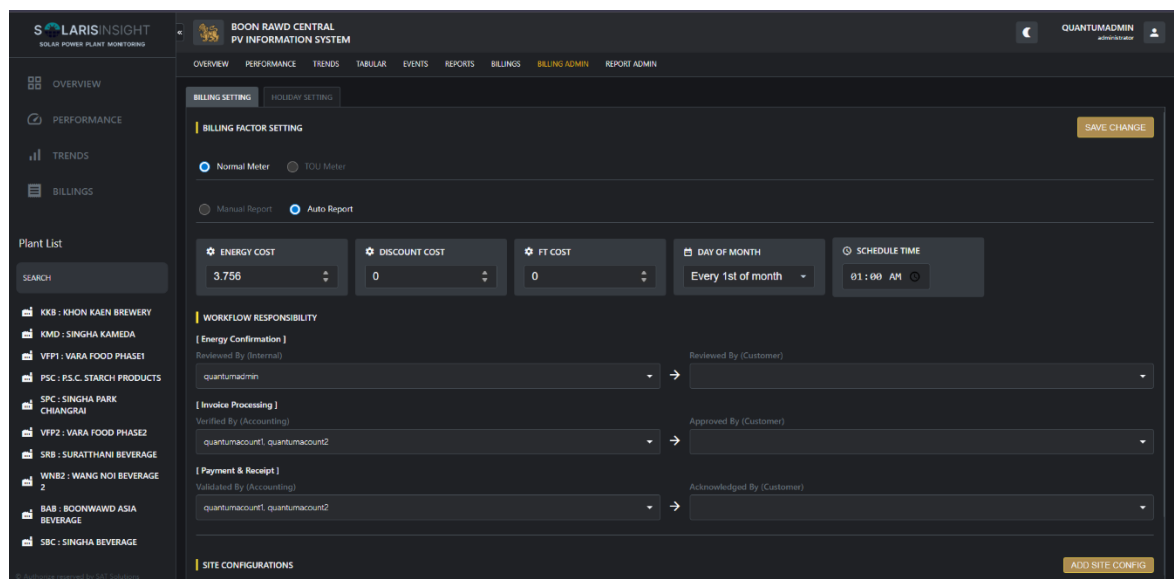
2.5 Report

Reports allow users to select report type, date range, and desired sites. A live preview of the report is displayed directly on the page. Users can review energy production, performance data, and statistics on a daily, monthly, or yearly basis. Reports can be downloaded using the **View** and **Download** buttons.



2.6 Billing Admin

The **Billing Admin** page provides a centralized interface for configuring billing parameters, scheduling billing generation, and managing workflow responsibilities across the billing lifecycle. It enables administrators to define both global and site-specific configurations to ensure accurate and consistent billing operations.



The screenshot displays the 'Billing Admin' page for Singha Kameda. The interface includes a sidebar with navigation options like Overview, Performance, Trends, and Billings. The main content area shows 'BILLING SETTINGS' and 'WORKFLOW RESPONSIBILITY'.

BILLING SETTINGS

- BILLING FACTOR SETTING**
 - ☒ Normal Meter ☐ ROU Meter
 - ☐ Manual Report ☒ Auto Report
 - ENERGY COST**: 3.756
 - DISCOUNT COST**: 0
 - FT COST**: 0
 - DAY OF MONTH**: Every 1st of month
 - SCHEDULE TIME**: 01:00 AM

WORKFLOW RESPONSIBILITY

- [Energy Confirmation]**
 - Reviewed By (Internal): quantumadmin
 - Reviewed By (Customer): [Dropdown]
- [Invoice Processing]**
 - Verified By (Accounting): quantumaccount1, quantumaccount2
 - Approved By (Customer): [Dropdown]
- [Payment & Receipt]**
 - Validated By (Accounting): quantumaccount1, quantumaccount2
 - Acknowledged By (Customer): [Dropdown]

SITE CONFIGURATIONS

Buttons: **SAVE CHANGE**, **ADD SITE CONFIG**

2.6..1 Billing Configuration

Users can configure key billing parameters that are applied during the billing generation process. The system supports multiple meter types and billing modes to accommodate different operational requirements.

Meter Type

- **Normal Meter:** Uses a single energy rate for all consumption.
- **TOU (Time of Use) Meter:** Supports different energy rates for On-Peak and Off-Peak periods.

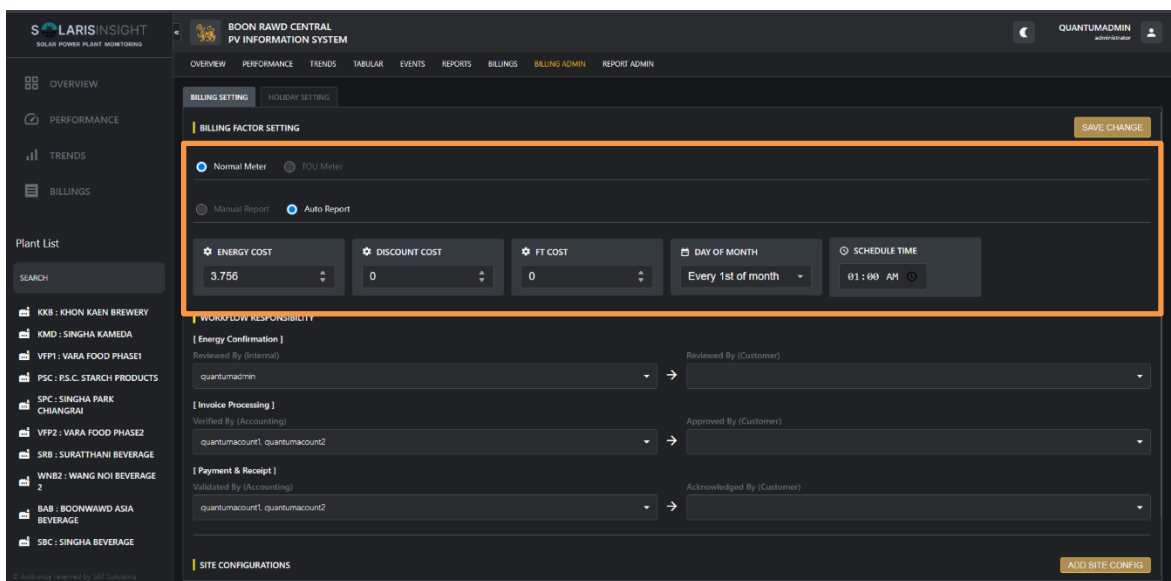
Billing Mode

- **Auto Report:** Billing is generated automatically based on a predefined schedule.
- **Manual Report:** Billing must be triggered manually by users.

Configurable Parameters

- Energy Cost (THB/kWh)
- Discount Cost (%)
- FT Cost (THB/kWh)
- Billing Generation Date (Day of Month)
- Schedule Time (Automatic execution time)

These parameters ensure that billing calculations are aligned with contractual and operational conditions.



2.6..2 Workflow Responsibility

The system allows administrators to define responsibility roles for each step of the billing workflow, ensuring clear accountability and structured approval processes.

Energy Confirmation

- Reviewed By (Internal)
- Reviewed By (Customer)

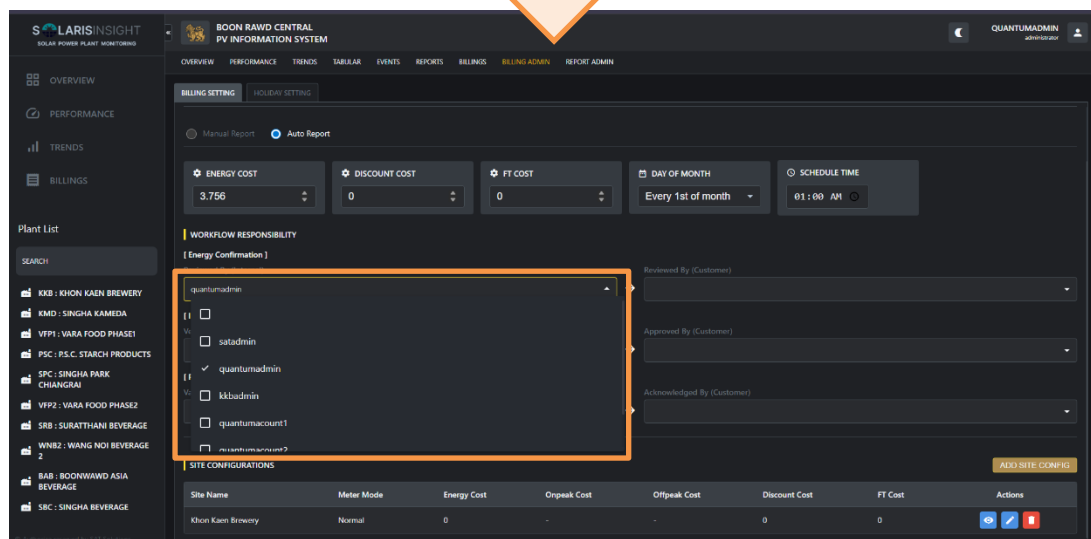
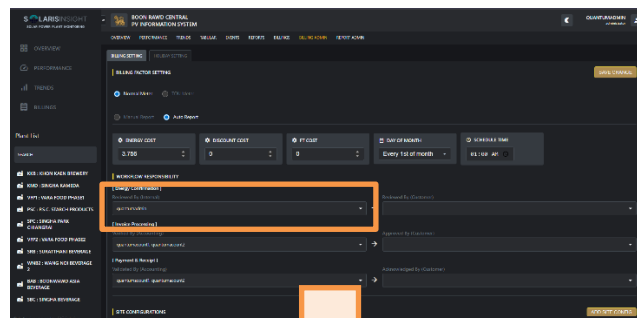
Invoice Processing

- Verified By (Accounting)
- Approved By (Customer)

Payment & Receipt

- Validated By (Accounting)
- Acknowledged By (Customer)

Each role can be assigned to specific users through selectable user lists. This structured workflow ensures that every billing record passes through verification, approval, and acknowledgment stages before completion.



2.6...3 Site Configuration

In addition to global settings, the system supports **site-specific billing configurations** to handle variations between different locations.

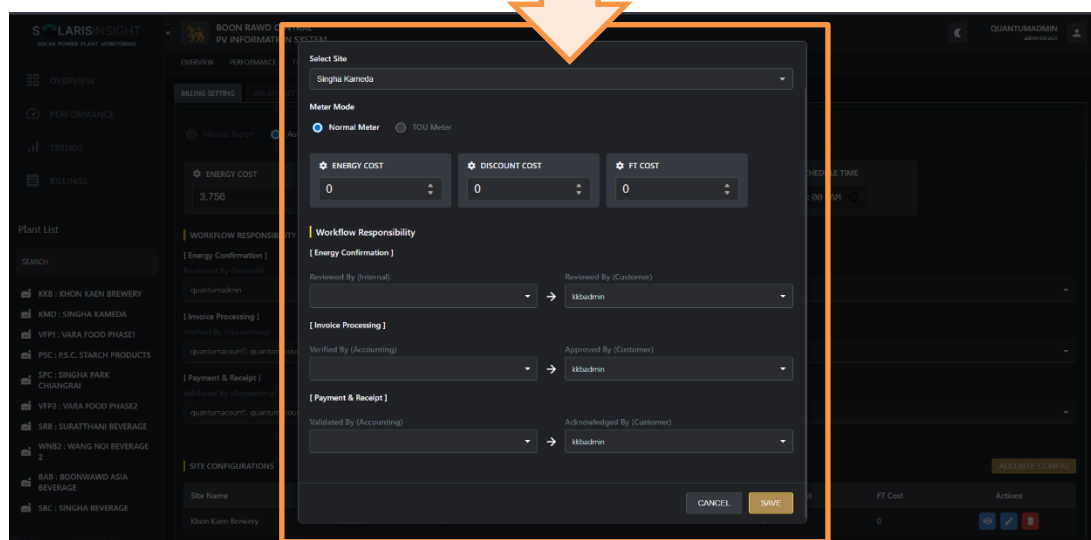
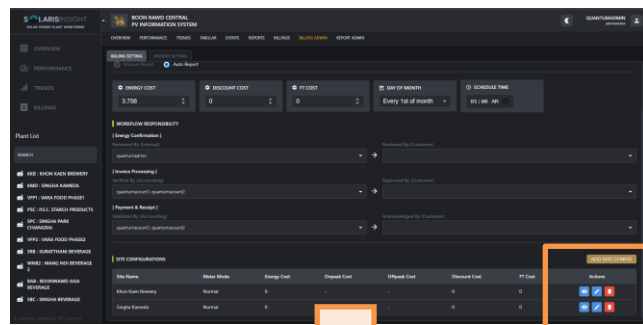
Users can:

- Add new site configurations via the **ADD SITE CONFIG** button
- Define meter type and billing parameters per site
- Assign workflow responsibilities specific to each site

Configured sites are displayed in a table, including:

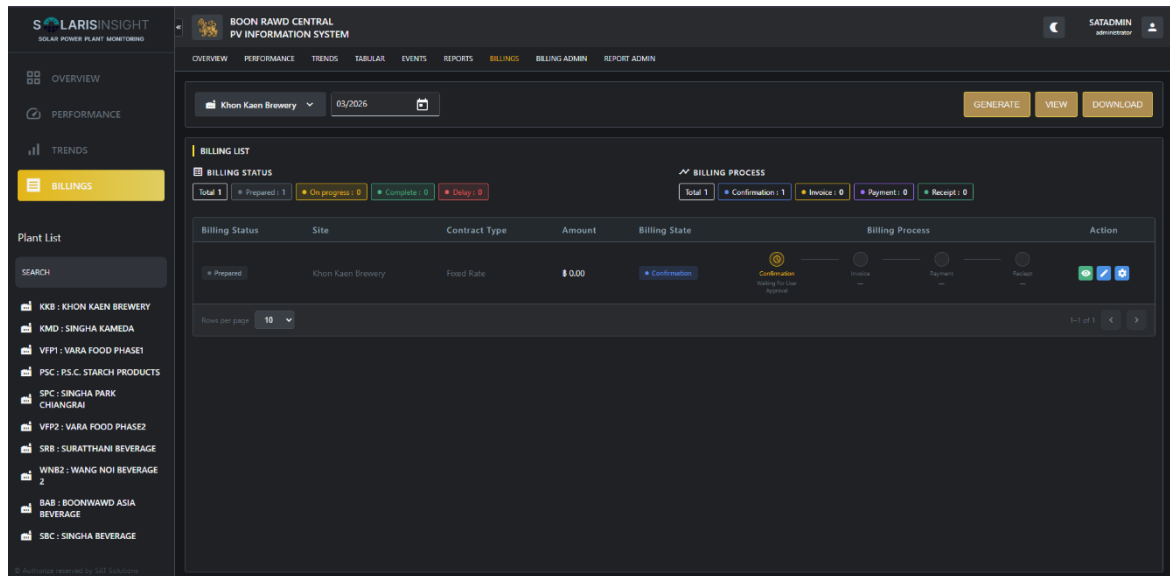
- Site Name
- Meter Mode
- Energy Cost / On-Peak / Off-Peak Rates
- Discount and FT Cost
- Action controls (View, Edit, Delete)

This flexibility allows the system to support diverse billing conditions across multiple sites within a single platform.



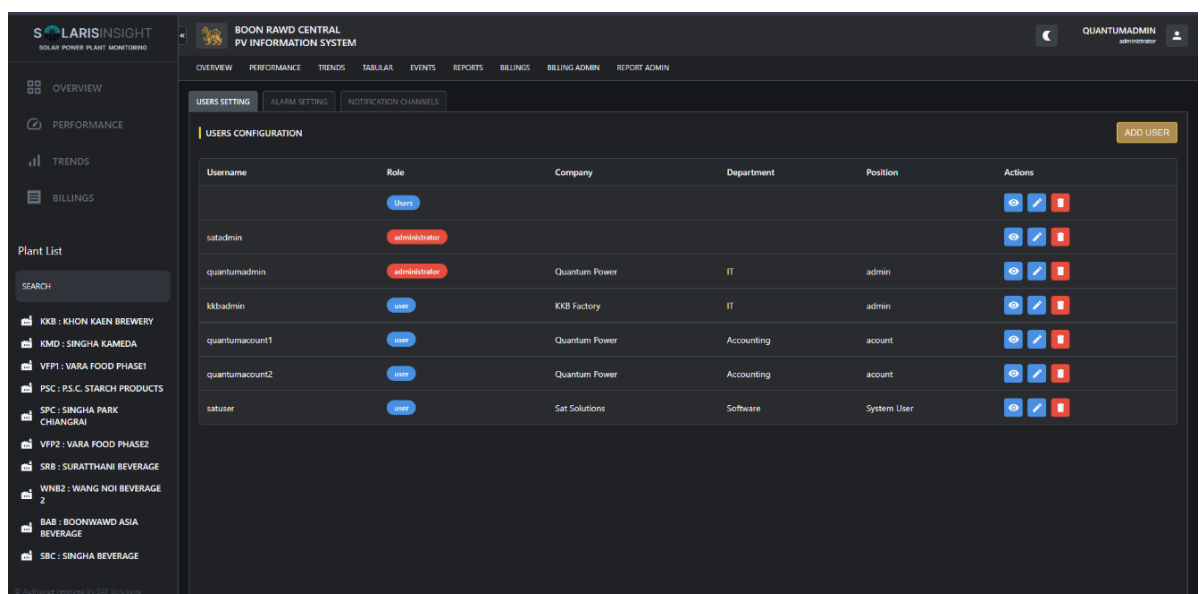
2.7 Billings

The **Billings** page provides a centralized interface for managing, tracking, and generating billing records across all sites. It enables users to monitor billing status, control the billing process, and access billing documents efficiently.



2.8 User Setting

The **User Setting** page provides a centralized interface for managing system users, roles, and access permissions. It enables administrators to control user access at both system and site levels, ensuring secure and structured operation across the platform.



2.8..1 User Management

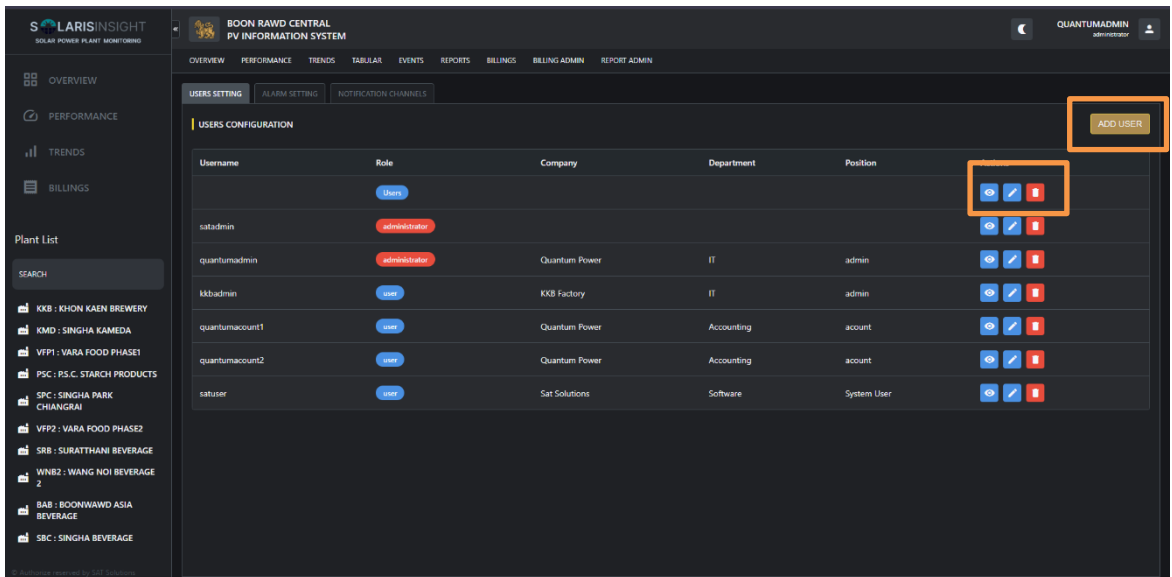
The system displays a list of all users in a structured table, including the following details:

- Username
- Role (Administrator / User)
- Company
- Department
- Position

Action Controls

- View user details
- Edit user information
- Delete user

Administrators can create new users using the **ADD USER** button, allowing full control over user onboarding.



The screenshot displays the 'SOLARISINSIGHT' interface for 'BOON RAWD CENTRAL PV INFORMATION SYSTEM'. The 'USERS CONFIGURATION' section shows a table of users. The 'ADD USER' button is highlighted in the top right corner.

Username	Role	Company	Department	Position	Action
satadmin	Administrator				View, Edit, Delete
quantumadmin	Administrator	Quantum Power	IT	admin	View, Edit, Delete
MMadmin	User	KKB Factory	IT	admin	View, Edit, Delete
quantumaccount1	User	Quantum Power	Accounting	account	View, Edit, Delete
quantumaccount2	User	Quantum Power	Accounting	account	View, Edit, Delete
satuser	User	Sat Solutions	Software	System User	View, Edit, Delete

2.8..2 User Configuration

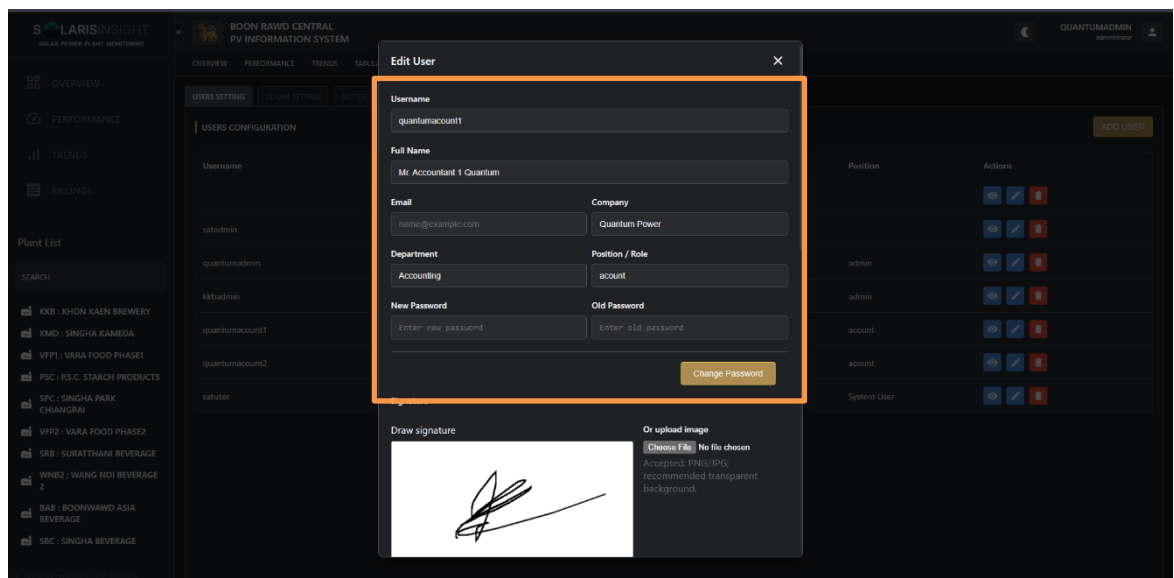
Each user profile can be configured with detailed information, including:

- Full Name
- Email Address
- Company
- Department
- Position / Role

Users can also update their credentials:

- Change password (with old password verification)

This ensures that user information remains accurate and up to date within the system.



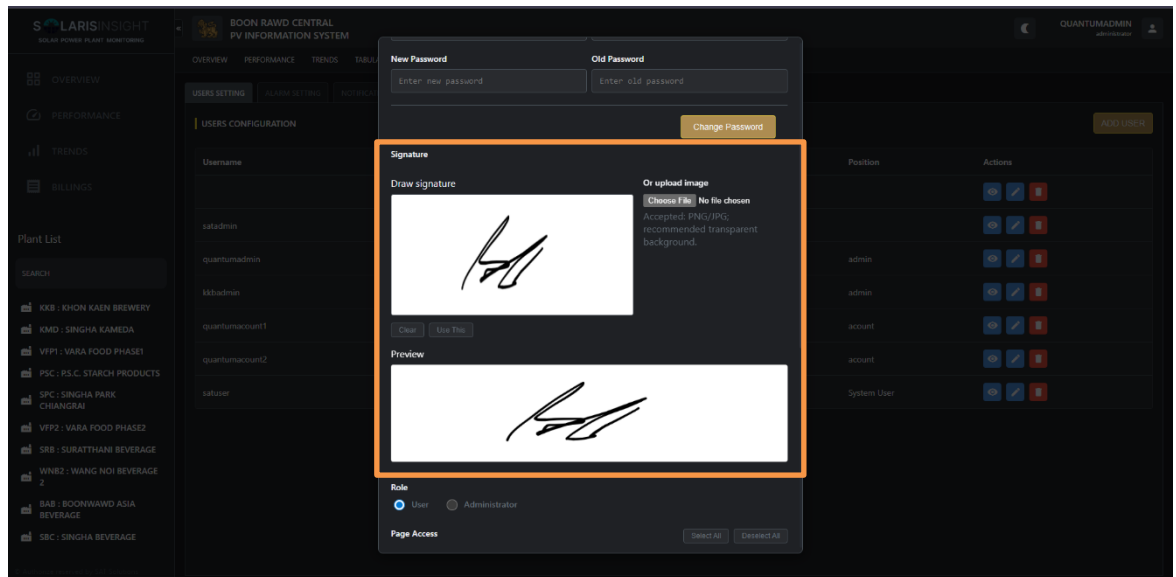
2.8..3 Signature Management

The system supports digital signature configuration for each user, which is used in billing and approval processes.

Users can:

- **Draw signature** directly within the system
- **Upload signature image** (PNG/JPG recommended with transparent background)
- Preview the signature before saving

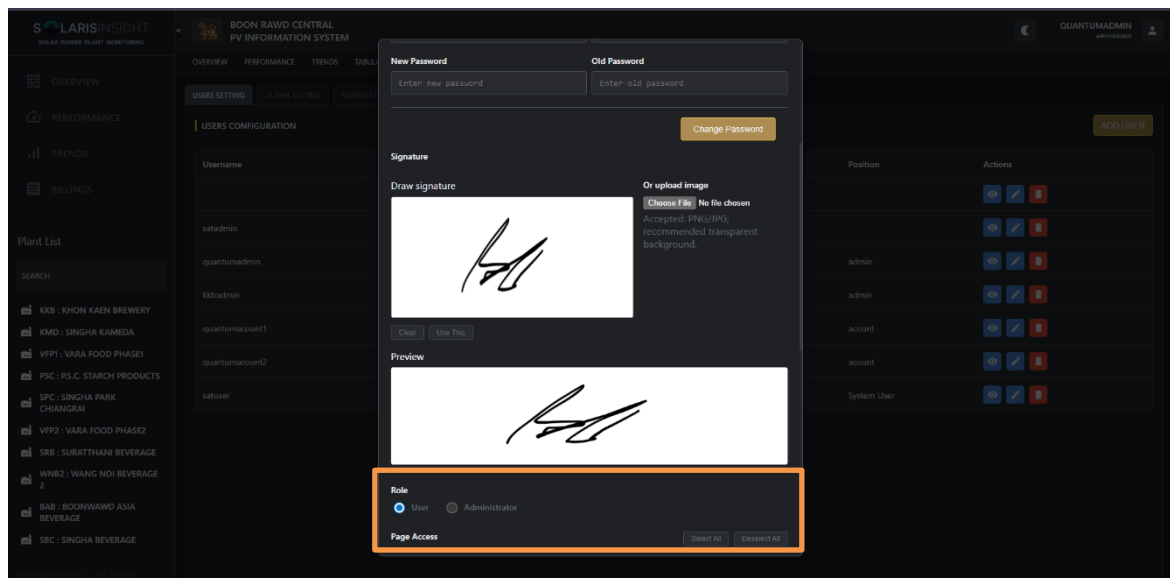
This feature enables seamless integration with document approval workflows.



2.8.4 Role & Permission Control

The system supports role-based access control (RBAC), allowing administrators to assign different levels of access:

- **Administrator:** Full access to all system features
- **User:** Restricted access based on assigned permissions





2.8..5 *Page Access Configuration*

Administrators can define access to specific system modules:

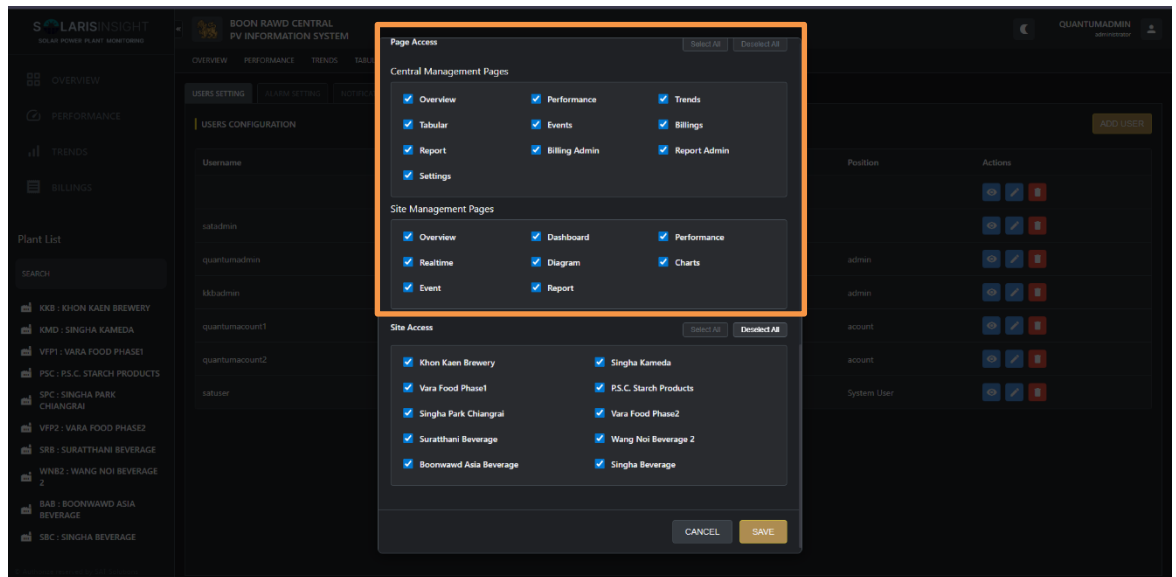
Central Management Pages

- Overview
- Performance
- Trends
- Tabular
- Events
- Billings
- Billing Admin
- Report Admin
- Settings

Site Management Pages

- Overview
- Dashboard
- Performance
- Realtime
- Diagram
- Charts
- Events
- Reports

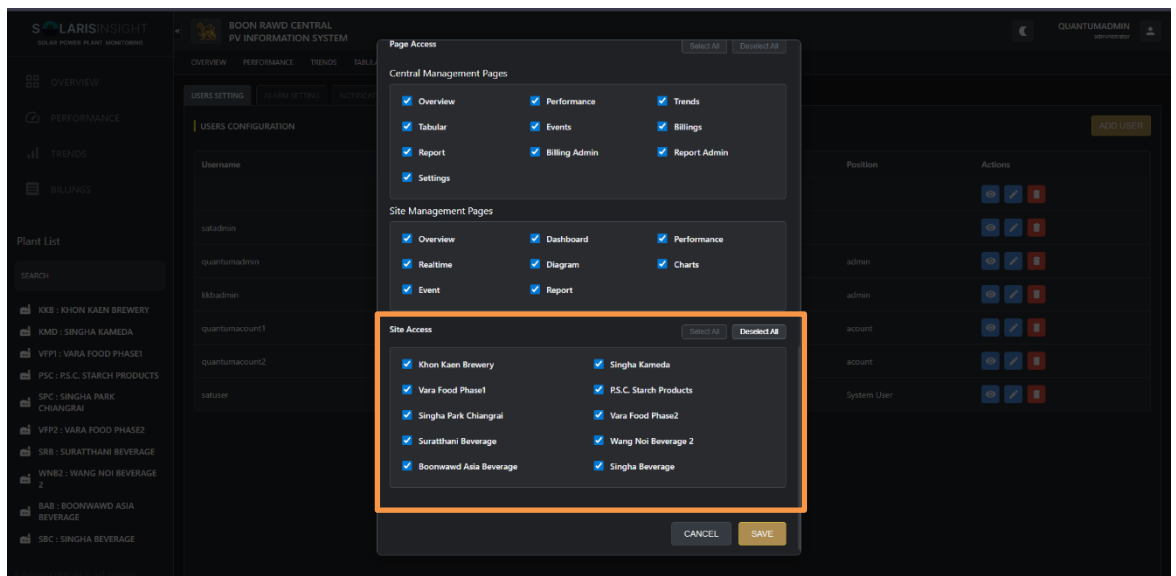
Users can be granted or restricted access on a per-page basis using selectable checkboxes.



2.8..6 Site Access Control

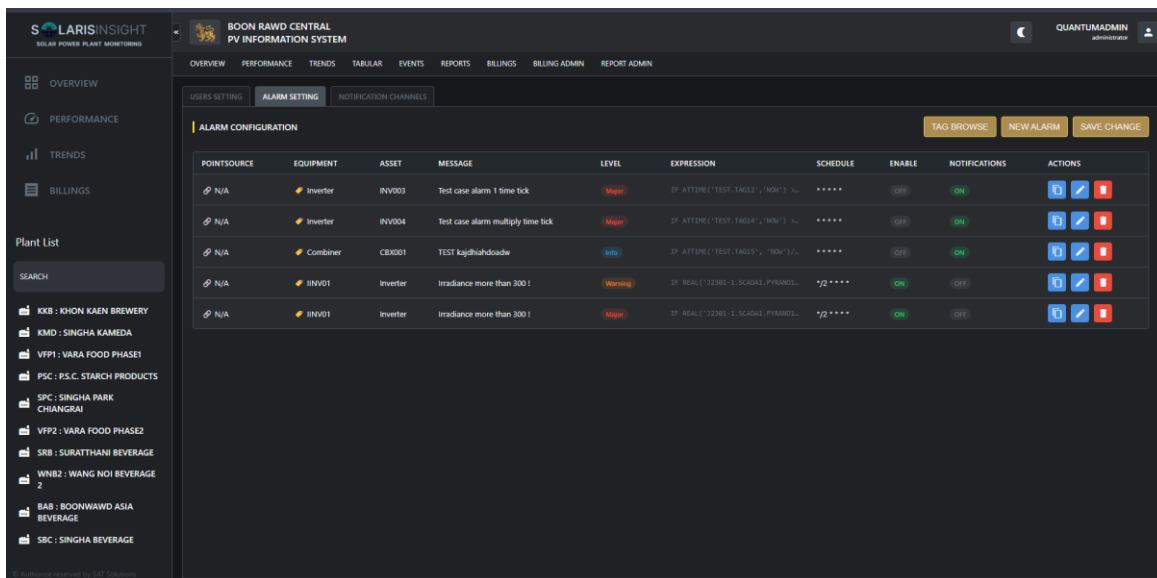
In addition to page-level permissions, administrators can define which sites each user can access.

- Multi-site selection supported
- “Select All” and “Deselect All” options available
- Ensures users only access relevant operational sites



2.9 Alarm Setting

The **Alarm Setting** page provides a centralized interface for configuring and managing system alerts across all monitored sites. It enables users to define alarm conditions, assign severity levels, and control notification behavior to ensure timely detection of abnormal events.



2.9.1 Alarm Creation & Configuration

Users can create new alarms using the **NEW ALARM** button. The configuration is divided into two main sections:

Basic Information

- Equipment (e.g., Inverter, Combiner)
- Asset (specific device identifier)
- Message (custom alarm description)
- Location (site reference)
- Point Source (data source such as SCADA)
- Level (Info / Warning / Major)
- Type (e.g., Alarm)

Trigger Configuration

Defines the logic used to trigger the alarm:

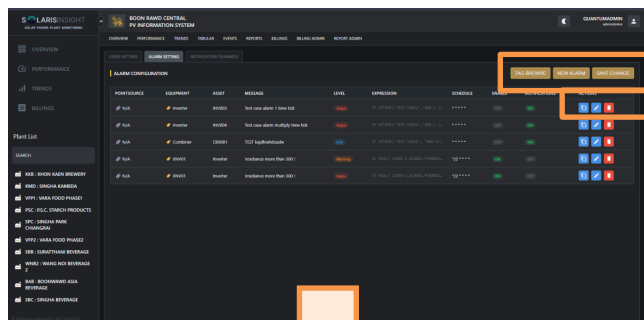
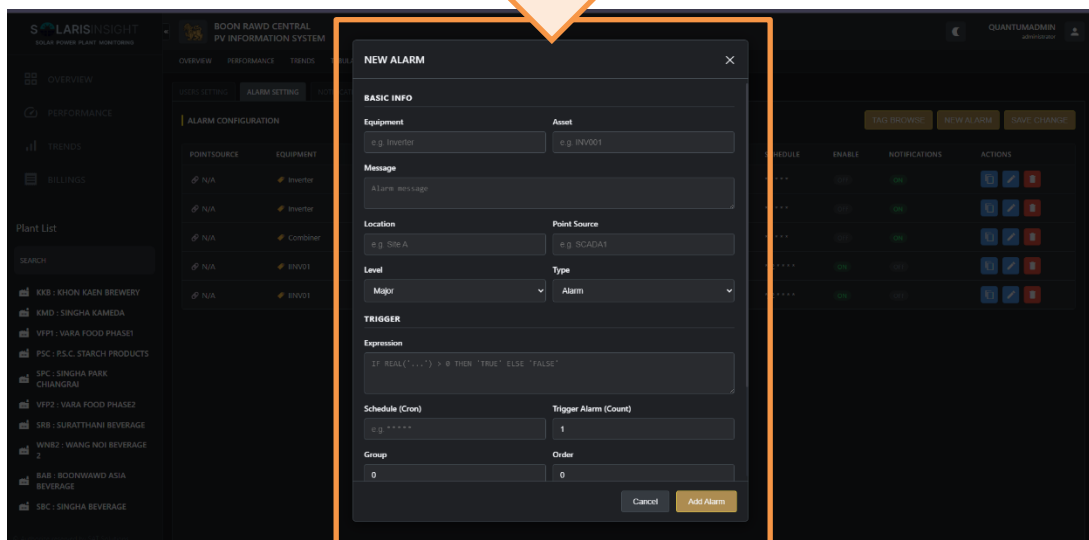
- **Expression**
Logical condition used to evaluate incoming data (e.g., threshold comparison, real-time value validation)
- **Schedule (Cron Format)**
Defines when the system evaluates the alarm condition

- **Trigger Alarm (Count)**
Number of occurrences required before triggering the alarm
- **Group / Order**
Used to control execution sequence and grouping of alarm conditions

This flexible configuration enables advanced rule-based monitoring aligned with real operational requirements.

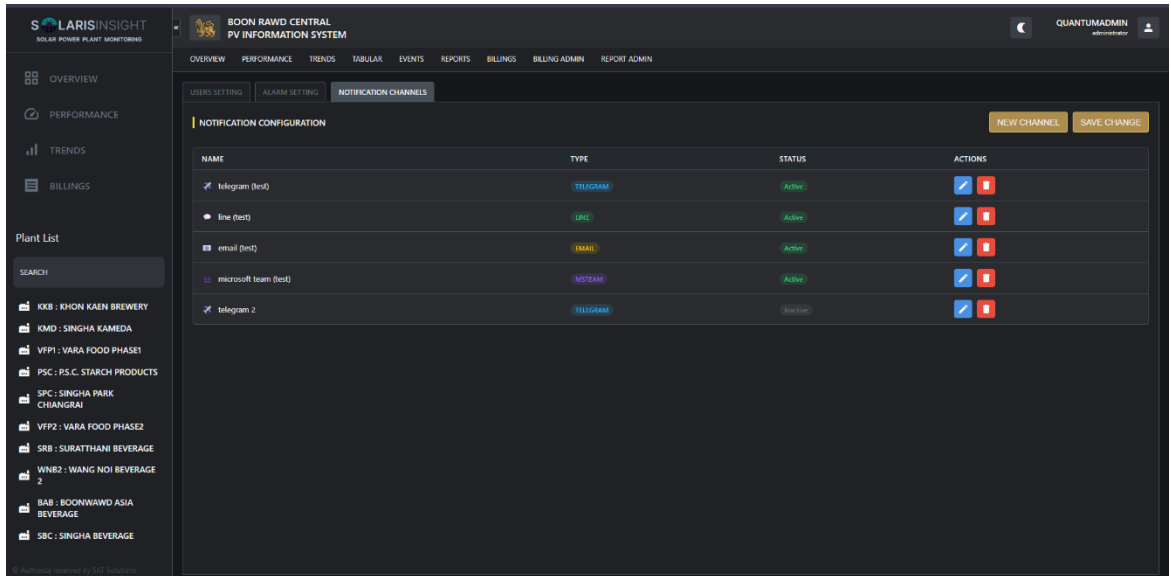
Alarm Control & Behavior

- **Enable / Disable Toggle**
Allows users to activate or deactivate specific alarms without deleting them
- **Notification Toggle**
Controls whether the alarm sends notifications when triggered
- **Tag Browse**
Assists users in selecting available data points for building expressions
- **Save Change**
Applies and persists all alarm configurations

2.10 Notifications Setting

The **Notification Channels** page provides a centralized interface for configuring and managing communication channels used for system alerts and notifications. It enables administrators to integrate multiple platforms and control how alarm events are delivered to users.



Notification Configuration Overview

The system displays all configured notification channels in a structured table, including:

- Channel Name
- Type (Telegram, LINE, Email, Microsoft Teams)
- Status (Active / Inactive)

Action Controls

- Edit channel configuration
- Delete channel

This overview allows administrators to manage all notification endpoints in one place.

Supported Notification Channels

The system supports multiple communication platforms to ensure flexibility and reliability in alert delivery:

- **Telegram**
- **LINE**
- **Email**



- **Microsoft Teams**

Each channel can be configured independently, allowing the system to support different communication preferences across users or organizations.

2.10..1 Channel Creation & Configuration

Users can create new notification channels using the **NEW CHANNEL** button. Configuration includes:

Basic Information

- Channel Name (custom identifier)
- Channel Type (select platform)
- Enable Channel (Active / Inactive toggle)

Channel Configuration

Depending on the selected channel type, users must provide required connection details:

- **Access Token**
Used for authentication with the selected platform (e.g., Telegram Bot Token)
- **User Group / Chat IDs**
Defines recipients or groups that will receive notifications

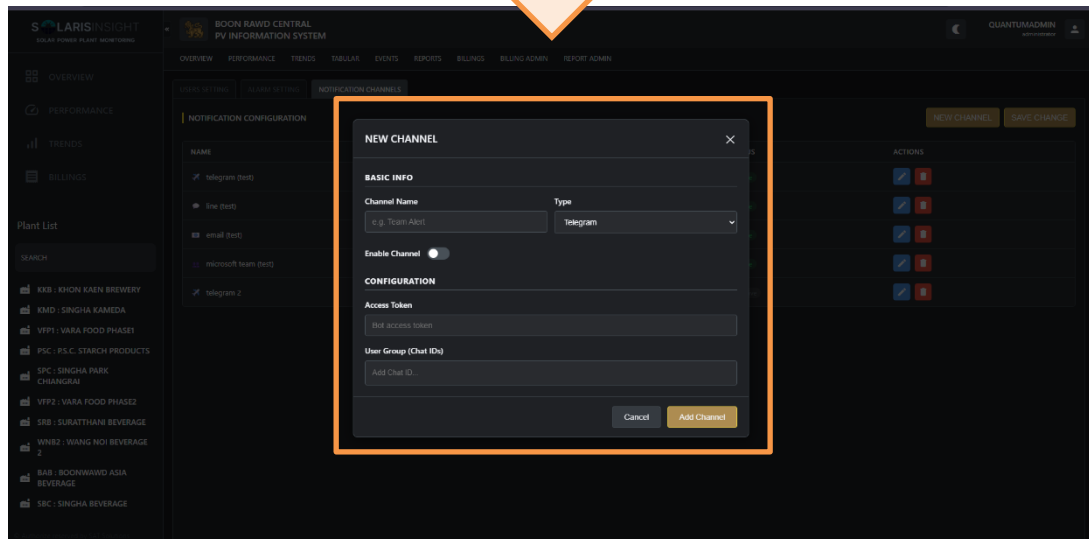
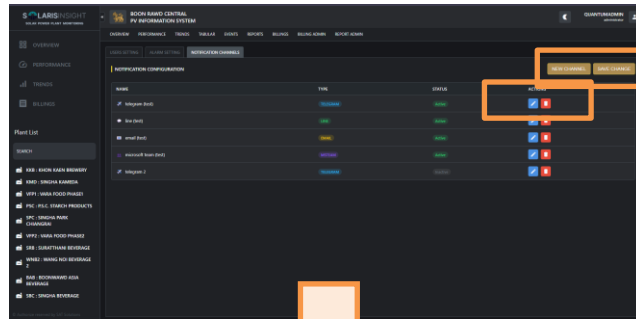
This configuration ensures that notifications are delivered to the correct audience.

Channel Behavior

- Only **active channels** are used for sending notifications
- Multiple channels can be configured and used simultaneously
- Channels can be modified or disabled without affecting existing alarm configurations
- Notification delivery is triggered by the **Alarm Setting** module

System Integration

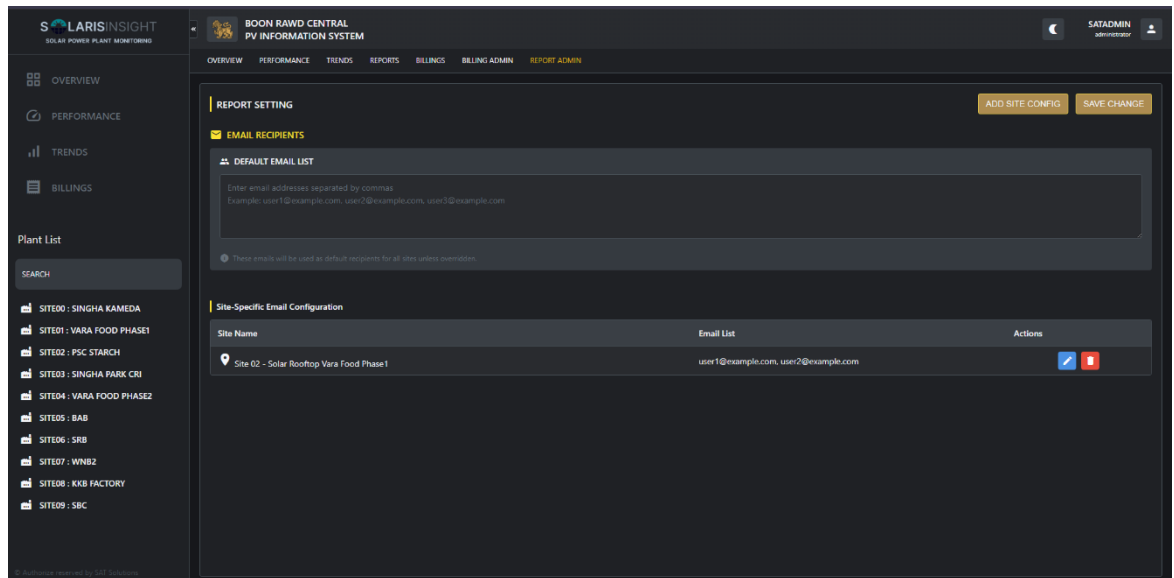
- Alarm events defined in the **Alarm Setting** module will trigger notifications through selected channels
- Each alarm can utilize one or more configured channels
- The system ensures reliable delivery across integrated platforms



2.11 Report Admin

The Report Admin allows configuration of report recipients. Users may set a **Default Email List** for all sites or configure **site-specific email lists**.

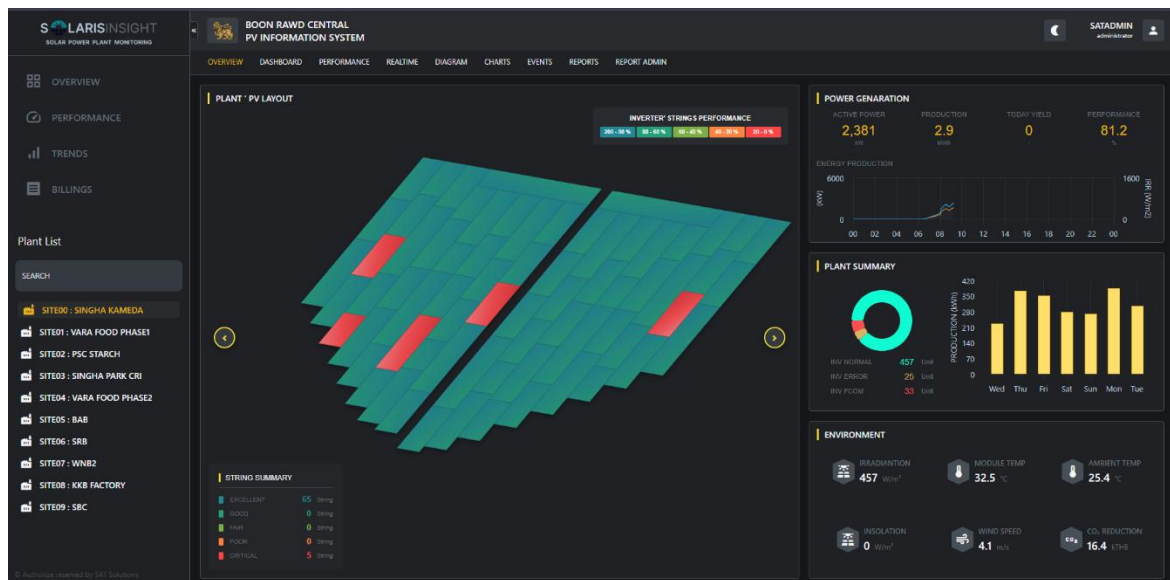
Functionality includes adding, editing, and removing email configurations, with a dedicated save button to confirm changes.



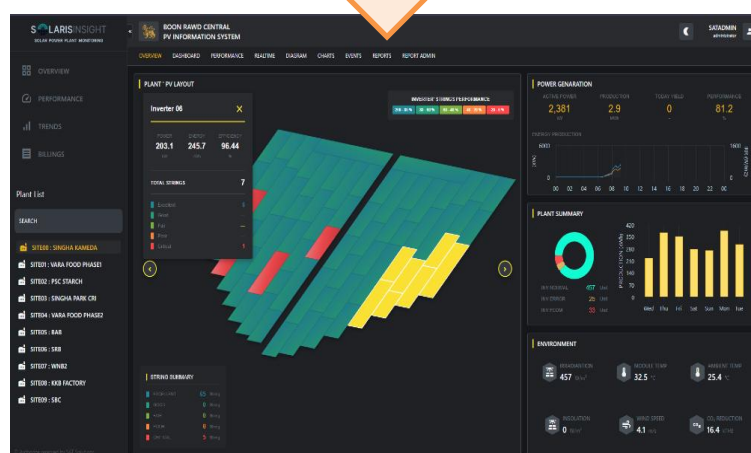
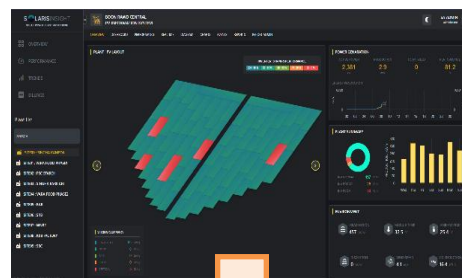
3 Site Monitoring Web Design

3.1 Site Overview

Displays the full layout of the solar plant, including real-time string-level status. It also summarizes key metrics such as Active Power, Production, and Yield, along with an hourly production chart. A factory status summary is displayed on the right for quick performance assessment.



String layout elements are clickable, allowing users to view inverter summary information.

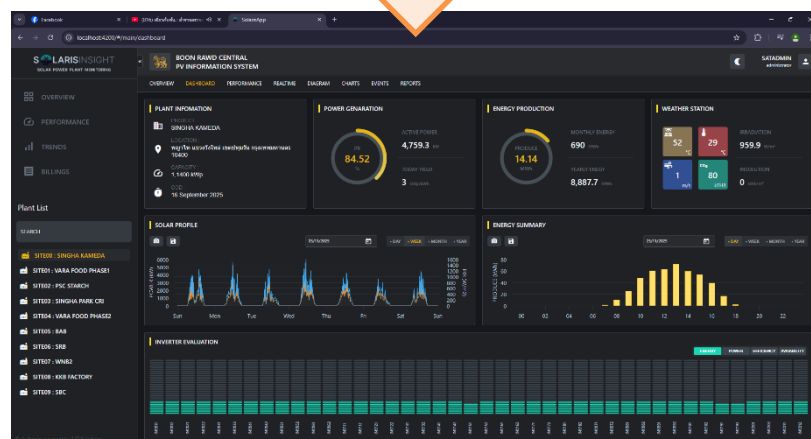
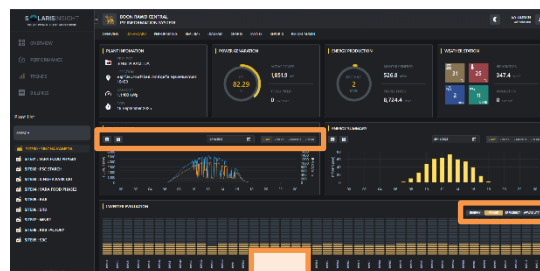


3.2 Dashboard

The Dashboard aggregates key plant information, including project overview, real-time power, and monthly/yearly energy production. It includes Solar Profile and Energy Summary charts for daily production insights and an Inverter Evaluation section comparing inverter performance.

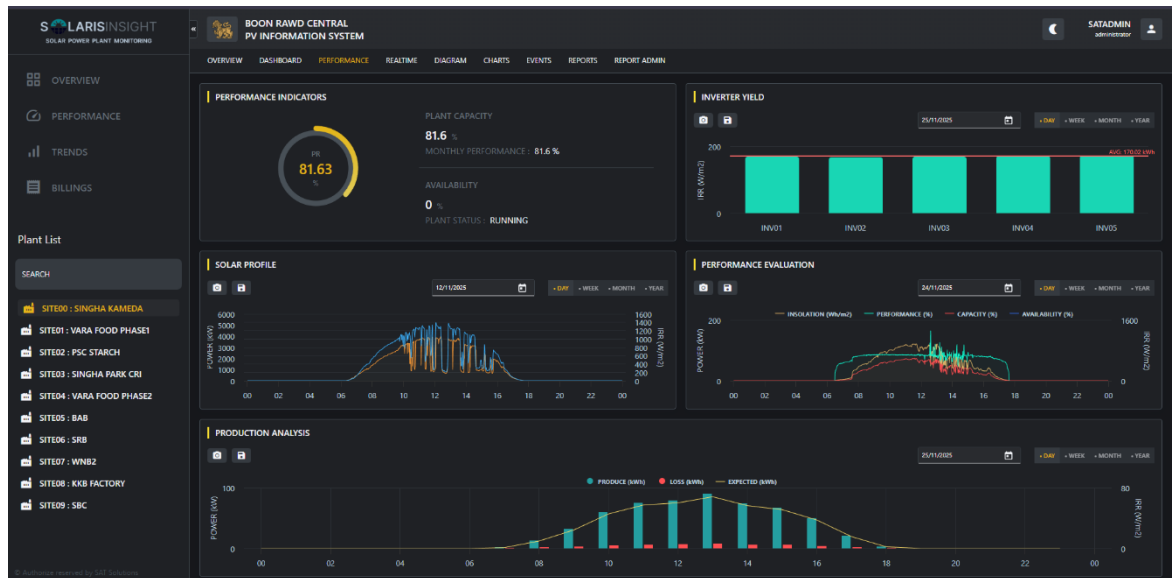


Users can select time ranges, download chart data in XLSX, or capture charts as PNG files.

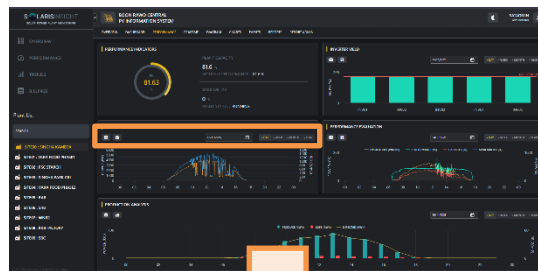


3.3 Performance

Displays essential performance indicators such as PR, Plant Capacity, and Availability. Charts include Inverter Yield and Solar Profile, along with Production Analysis for comparing actual vs. expected production, helping identify system losses.

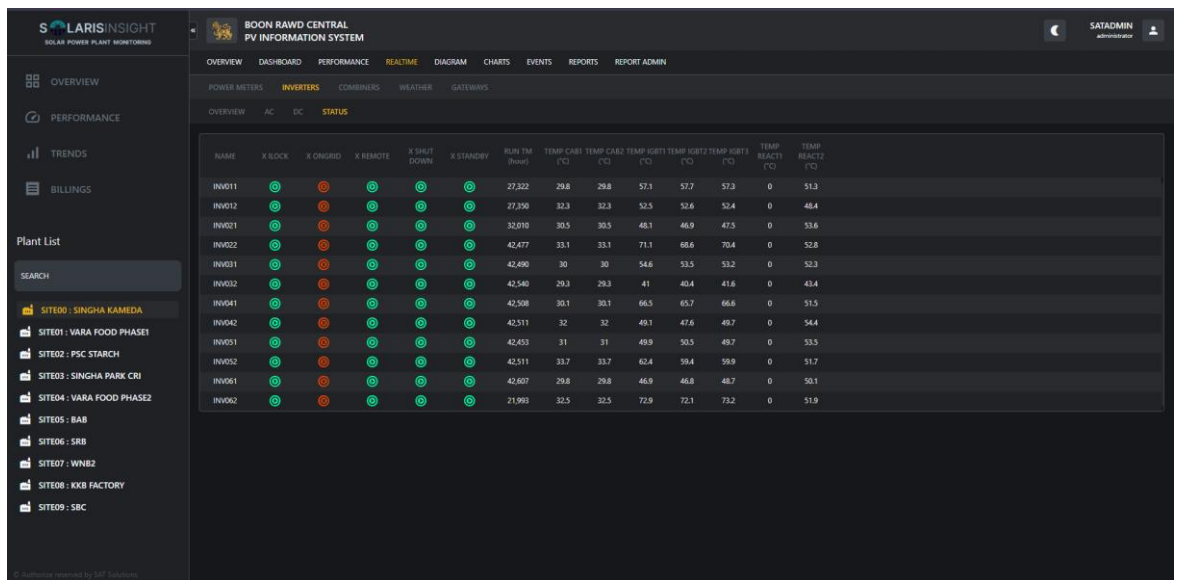


Users can select time ranges, download chart data in XLSX, or capture charts as PNG files.



3.4 Real-time

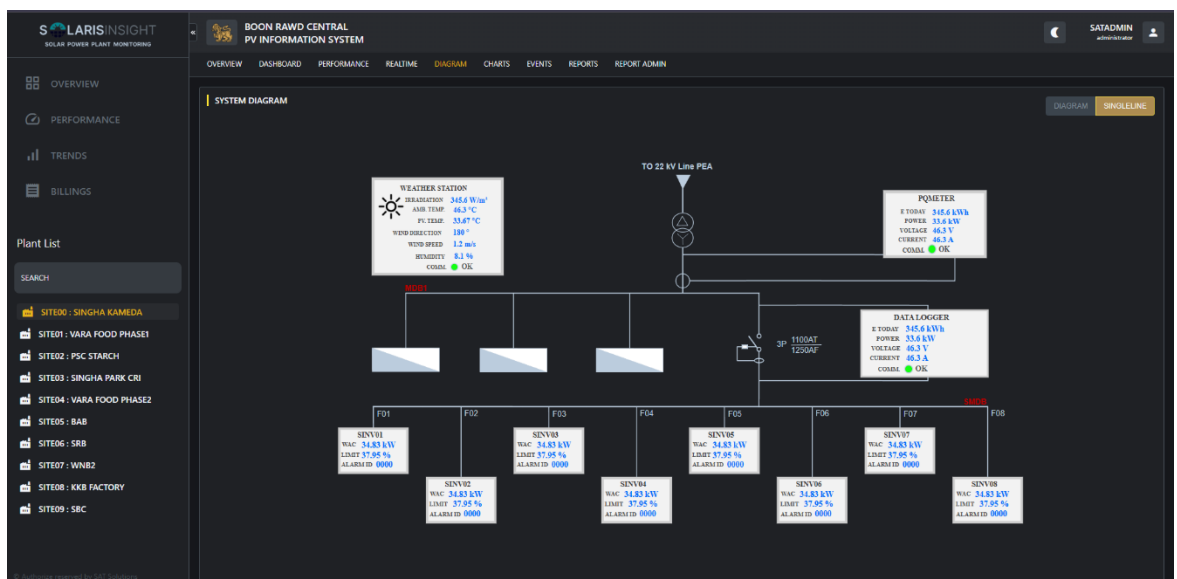
Provides real-time operational status of site devices, including inverters, sensors, and meters. Users can review key operational metrics such as temperature, voltage, current, and operating hours to detect abnormalities early.



NAME	X BLOCK	X ONGRID	X REMOTE	X SHUT DOWN	X STANDBY	RUN TM (hours)	TEMP CAB1 (°C)	TEMP CAB2 (°C)	TEMP KBT1 (°C)	TEMP KBT2 (°C)	TEMP KBT3 (°C)	TEMP REACT1 (°C)	TEMP REACT2 (°C)
INV011						27,322	29.8	29.8	57.1	57.7	57.3	0	51.3
INV012						27,350	32.3	32.3	52.5	52.6	52.4	0	48.4
INV021						32,010	30.5	30.5	46.1	46.9	47.5	0	53.6
INV022						42,477	33.1	33.1	71.1	68.6	70.4	0	52.8
INV031						42,490	30	30	54.6	53.5	53.2	0	52.3
INV032						42,540	29.3	29.3	41	40.4	41.6	0	43.4
INV041						42,508	30.1	30.1	66.5	65.7	66.6	0	51.5
INV042						42,511	32	32	49.1	47.6	49.7	0	54.4
INV051						42,453	31	31	49.9	50.5	49.7	0	53.5
INV052						42,511	33.7	33.7	62.4	59.4	59.9	0	51.7
INV061						42,607	29.8	29.8	46.9	46.8	48.7	0	50.1
INV062						21,993	32.5	32.5	72.9	72.1	73.2	0	51.9

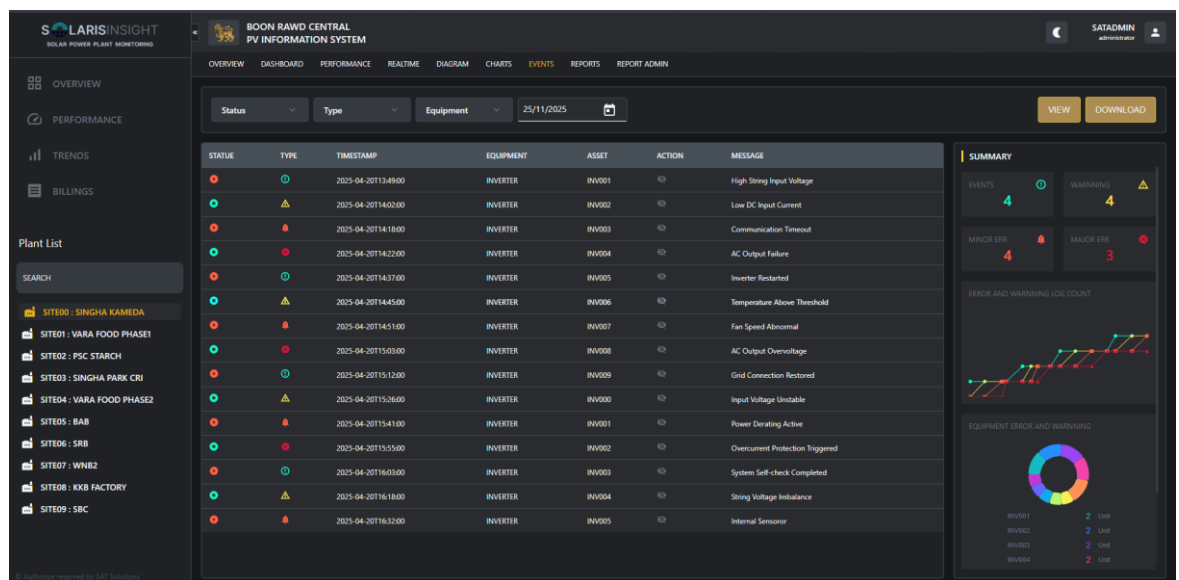
3.5 Diagram

Shows a complete system diagram, covering Weather Station, Meters, Data Logger, and Inverters. It provides a centralized view of device connectivity and operational flow throughout the plant.

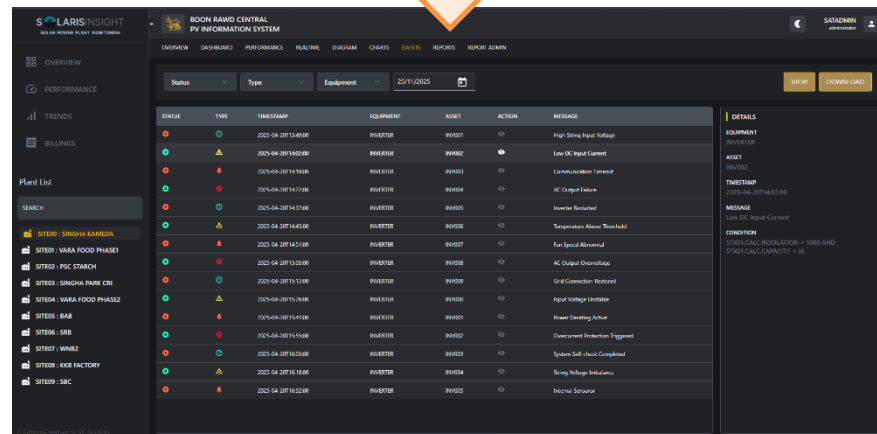
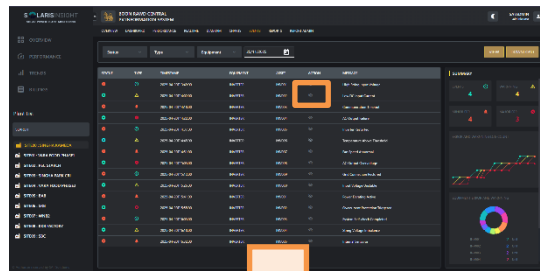


3.6 Events

Displays all recorded events and alerts. Users can filter events by status, device type, or date. Details such as affected equipment, alert messages, and timestamps are clearly shown. Charts on the right present event summaries, trend analysis, and distribution of Errors/Warnings.

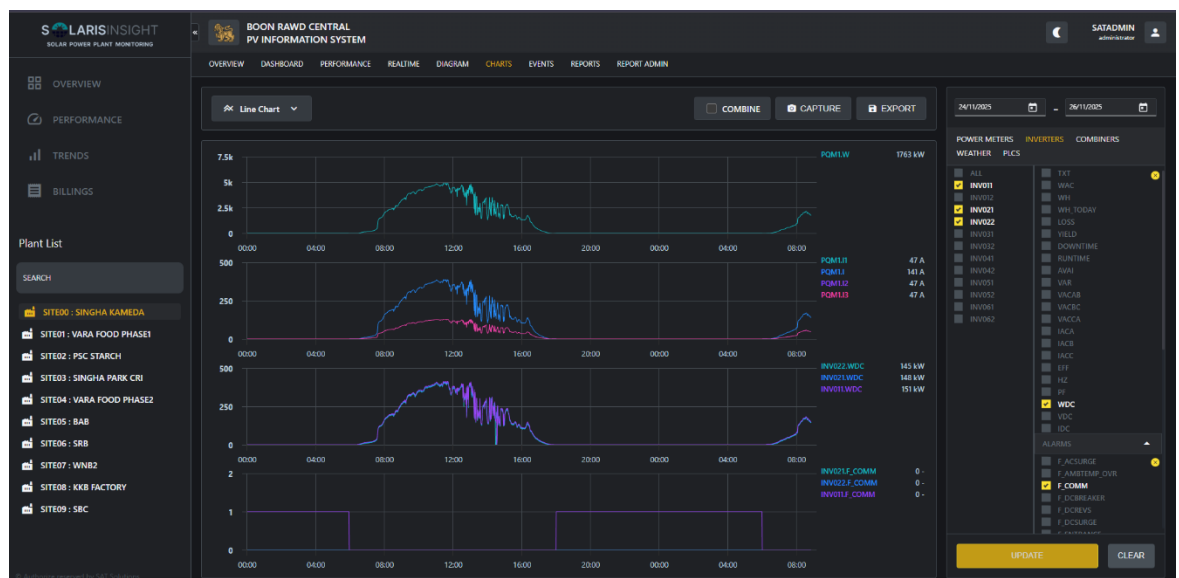


Detailed event information can be viewed by clicking the eye icon.

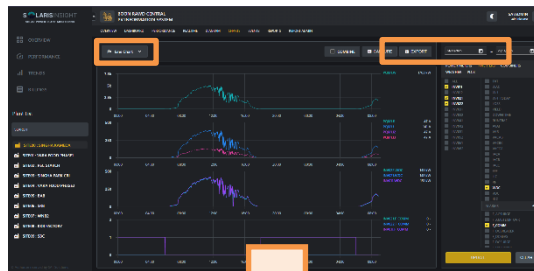


3.7 Charts

Provides advanced charting tools for comparing parameters across various device categories (Inverter, Weather, Meter).



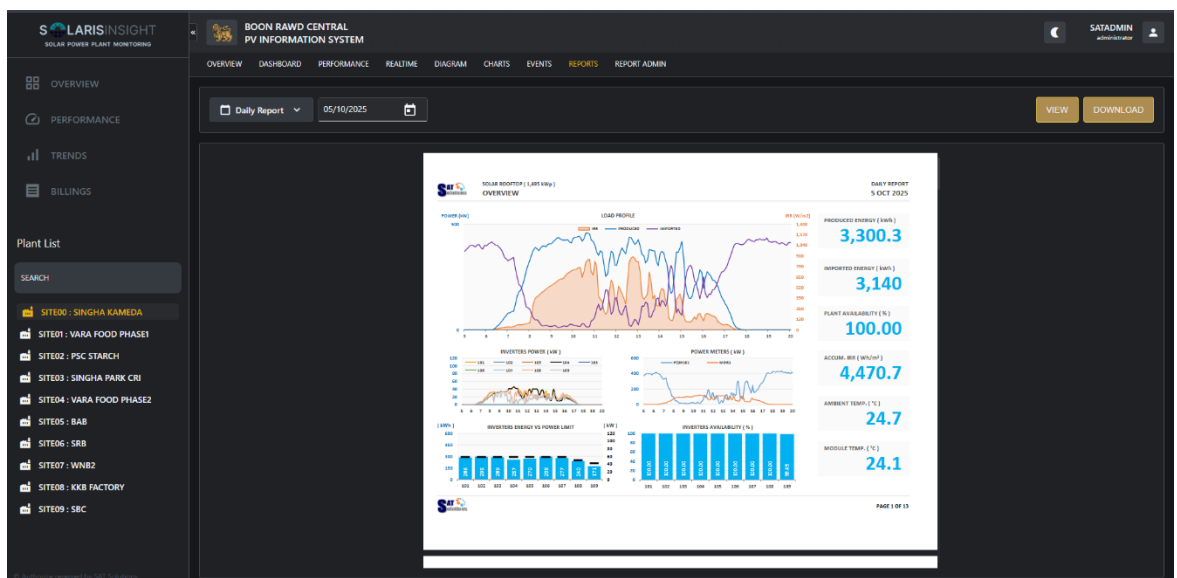
Users can combine multiple charts, adjust chart formats, and export data as PNG or XLSX.



3.8 Reports

Reports Allows users to select report type and date ranges for the site and displays a real-time preview.

Reports can be downloaded using **View** or **Download** buttons for convenient access.





4.2 Billing Dashboard

System The Billing Dashboard provides a centralized interface to monitor and manage all billing activities.

Key Components

Billing Status Summary

- Prepared
- On Progress
- Complete
- Delay

Billing Process Tracker

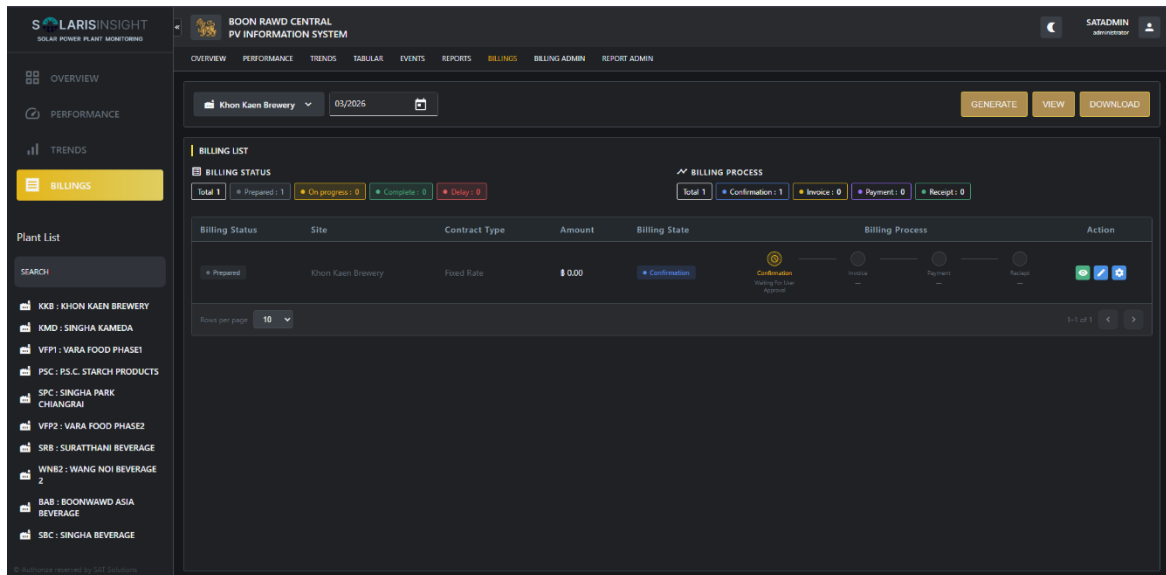
- Confirmation
- Invoice
- Payment
- Receipt

Billing List Table

- Site
- Contract Type
- Amount
- Billing State
- Process Progress
- Actions

User Actions

- Generate billing data
- View documents
- Download reports
- Execute workflow actions



4.3 Energy Confirmation (Step 1)

This step validates energy data before initiating billing.

Process

- System generates the **Confirmation Energy document**
- Internal team performs data verification
- Customer reviews and confirms

Features

- PDF preview with zoom and navigation tools
- Approval / rejection controls
- Activity log tracking (user & timestamp)
- Optional comment submission

4.3..1 Internal team data verification

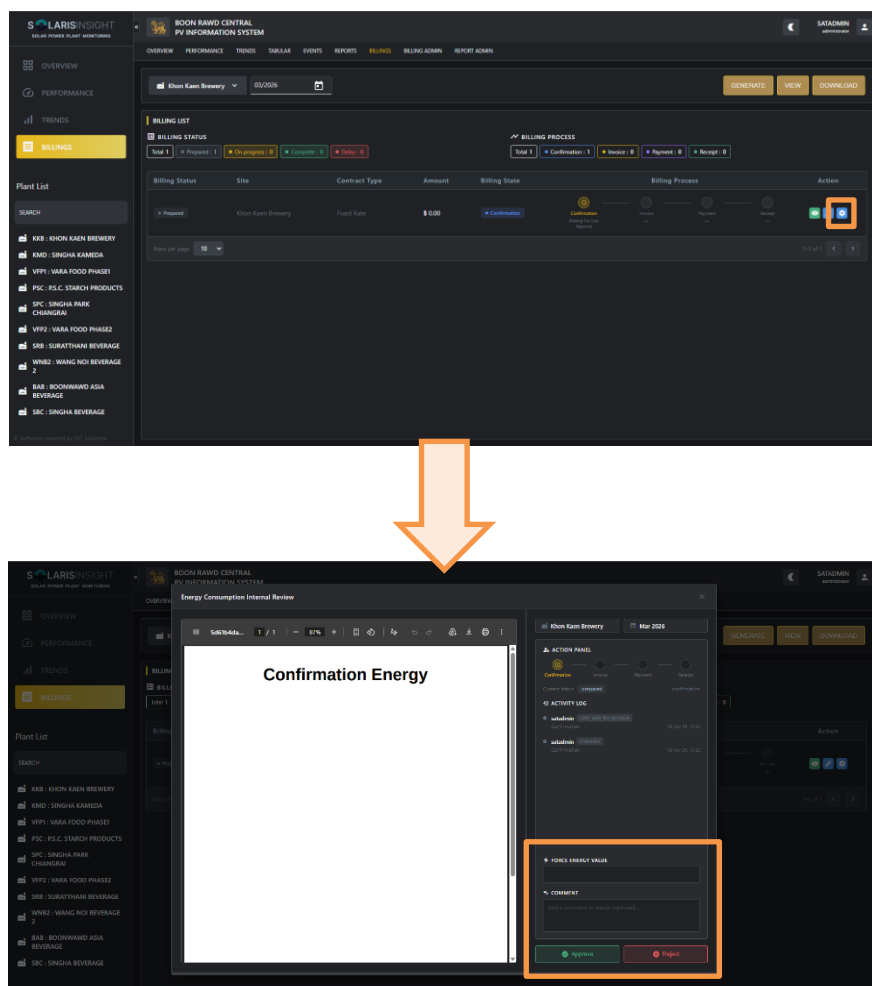
This step ensures that energy data is validated before customer confirmation.

Process

1. Go to **Billings** → select Site and billing period
2. Open the billing record via the **Action button**
3. Review the **Confirmation Energy PDF**
4. Verify:
 - Total energy generation
 - Meter readings and calculations
 - Data completeness and anomalies
5. (Optional) Add **Force Energy Value** or comments

Outcome

- **Approve** → Proceed to customer confirmation
- **Reject** → Return for correction



4.3..1.1 Internal team reject

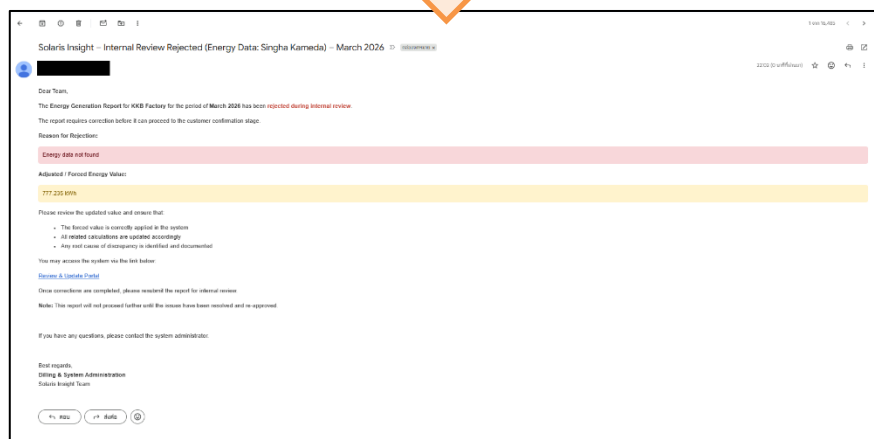
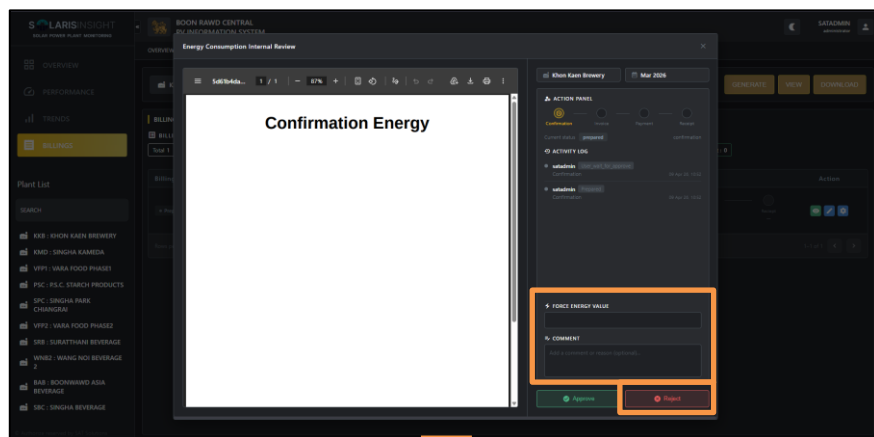
This step occurs when the internal team identifies incorrect or incomplete energy data during validation.

Process

- Reviewer clicks **Reject** in the Energy Confirmation review window
- Provide **reason for rejection** in the comment field
- (Optional) Input **adjusted / forced energy value**
- Submit the rejection

Outcome

- System updates status to **Rejected (Internal Review)**
- Notification email is sent to relevant users
- Data is returned for correction before re-submission



4.3..2 Customer reviews and confirms

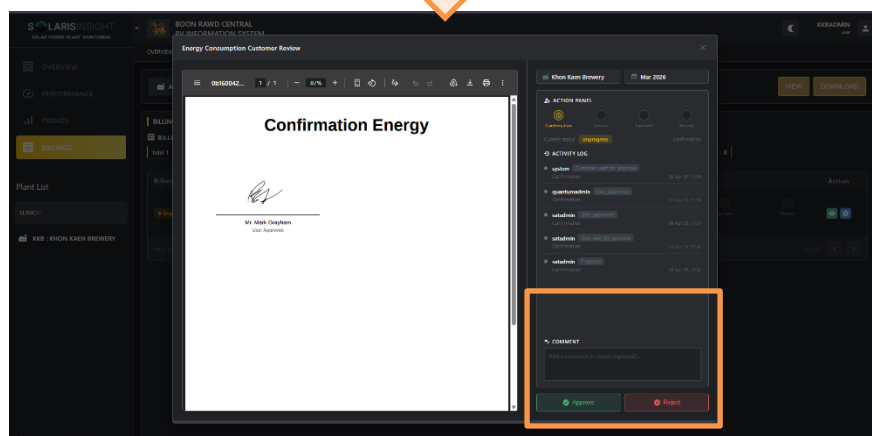
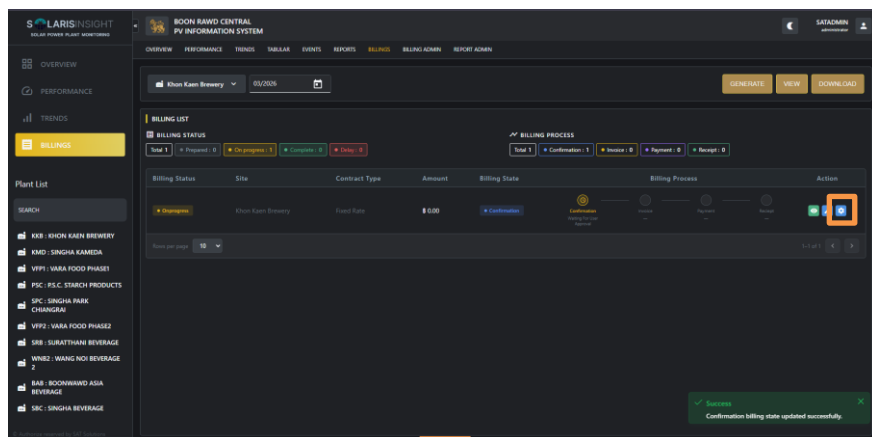
This step allows the customer to review and confirm the validated energy data.

Process

1. Open the billing record via the **Action** button
2. Review the **Confirmation Energy PDF**
3. Verify energy data and document details
4. (Optional) Add comments

Outcome

- **Approve** → Proceed to invoice generation
- **Reject** → Return for internal revision



4.3..2.1 Customer reject

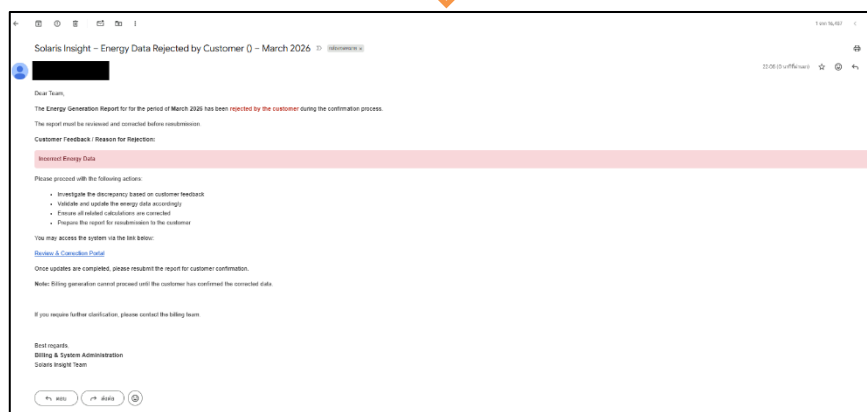
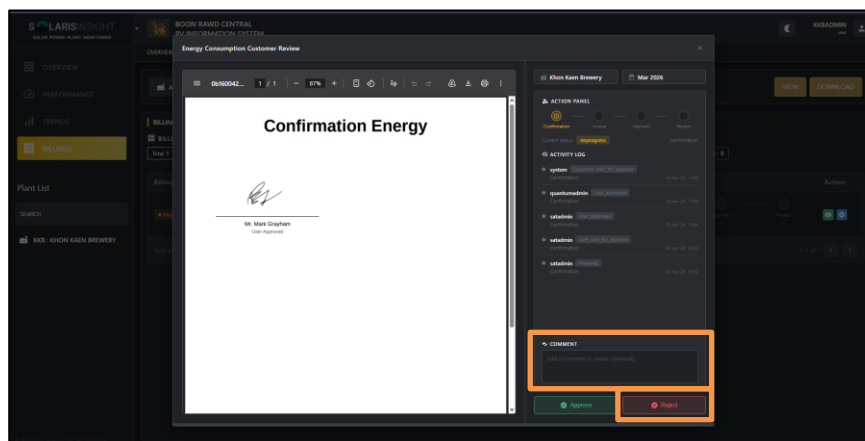
This step occurs when the customer identifies issues in the energy data during confirmation.

Process

- Customer clicks **Reject** in the Confirmation Energy review window
- Provide **reason for rejection** in the comment field
- Submit the rejection

Outcome

- System updates status to **Rejected (Customer Review)**
- Notification email is sent to internal team
- Data is returned for correction and re-validation





4.4 Invoice Management (Step 2)

Handles invoice generation and approval workflow.

Process

- Generate invoice document
- Internal accounting approval
- Customer acknowledgment

Features

- Digital approval workflow
- Signature tracking
- Status transition (Prepared → Approved)
- Full audit trail

4.4.1 *Internal accounting approval*

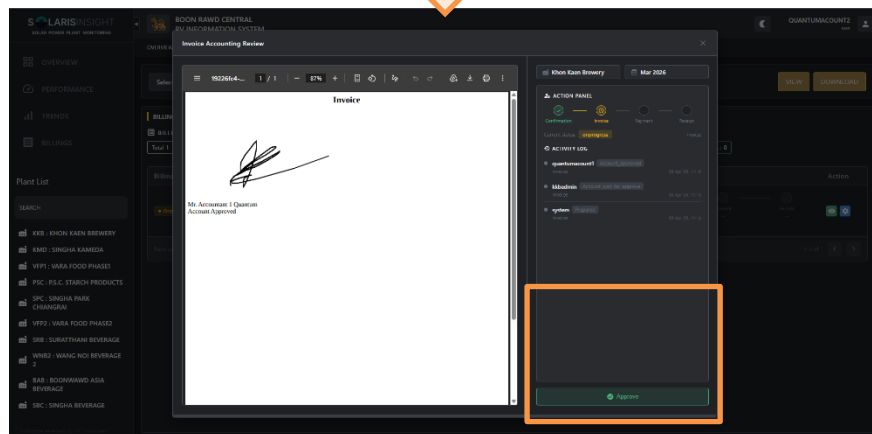
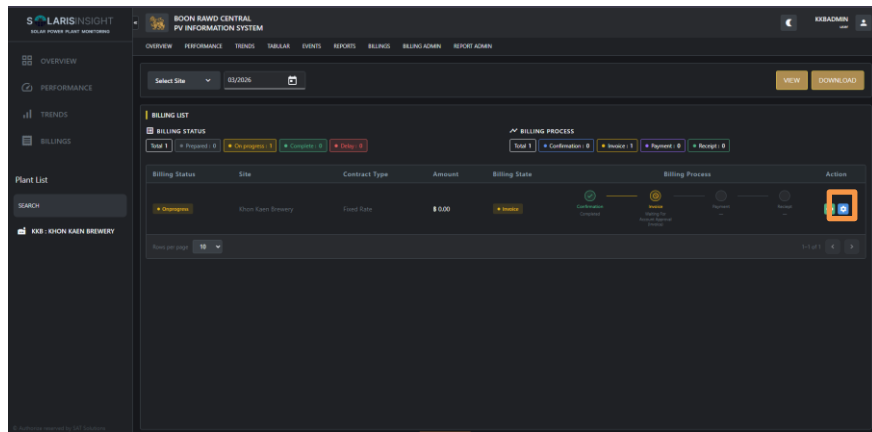
This step ensures the invoice is reviewed and approved by the internal accounting team before sending to the customer.

Process

1. Open the billing record via the **Action button**
2. Review the **Invoice document (PDF)**
3. Verify:
 - Billing calculations and tariff
 - Total payable amount
 - Supporting data consistency
4. (Optional) Add comments

Outcome

- **Approve** → Proceed to customer invoice review
- **Reject** → Return for correction





4.5 Signed Invoice Submission

Customers must return a signed invoice for validation.

Capabilities

- Upload signed invoice (PDF)
- Specify document submission date
- File format validation

This ensures legal and financial compliance before payment processing.

4.5.1 *Customer acknowledgment*

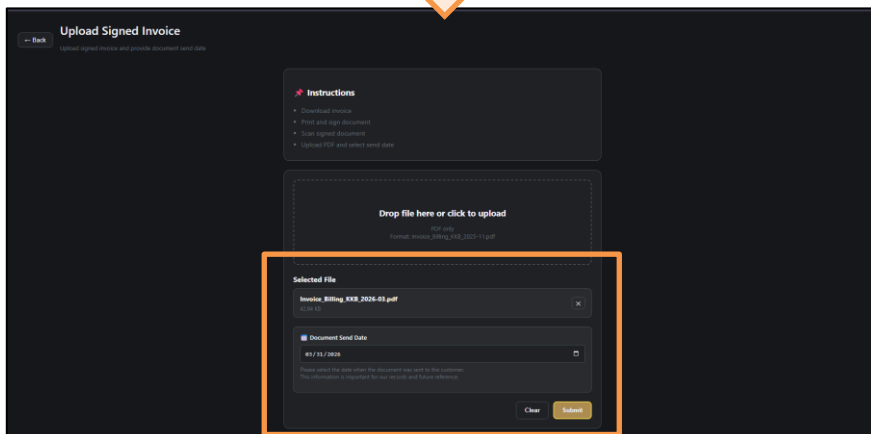
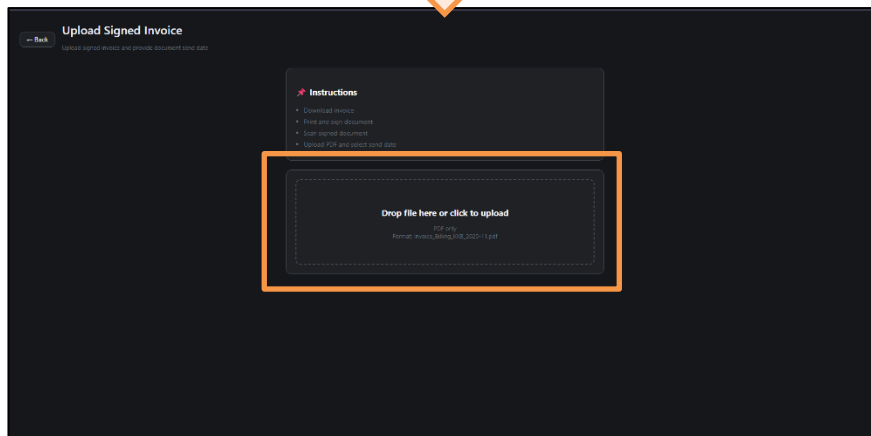
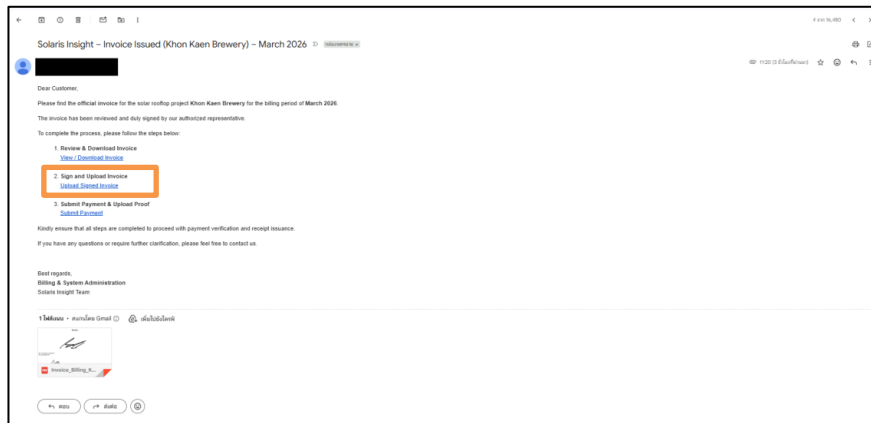
This step allows the customer to review, sign, and upload the approved invoice.

Process

1. Access the invoice via the provided link
2. Download and review the **Invoice document**
3. Sign the invoice
4. Upload the signed document via the upload page
5. Enter the **document send date** and submit

Outcome

- Signed invoice is submitted for payment process
- System updates status to **Ready for Payment**





4.6 Payment Submission (Step 3)

This step allows the customer to submit payment details and proof of transfer.

Process

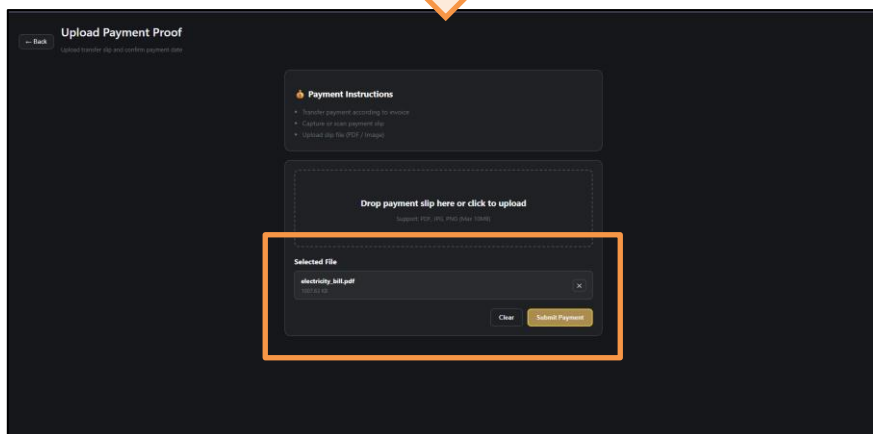
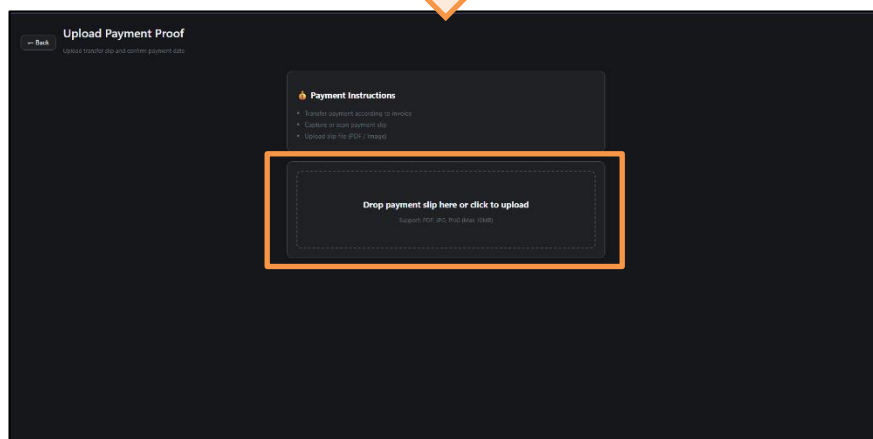
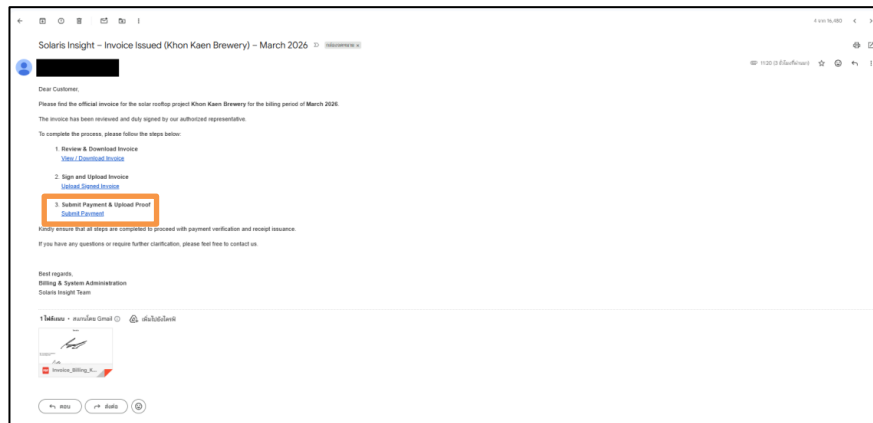
1. Access the payment page via the provided link
2. Upload the **payment proof** (PDF / Image)
3. Review uploaded file details
4. Submit payment

Features

- Drag & drop file upload
- Multi-format support
- Payment tracking and status monitoring

Outcome

- Payment is submitted for verification
- System updates status to **Pending Verification**



4.7 Payment Verification

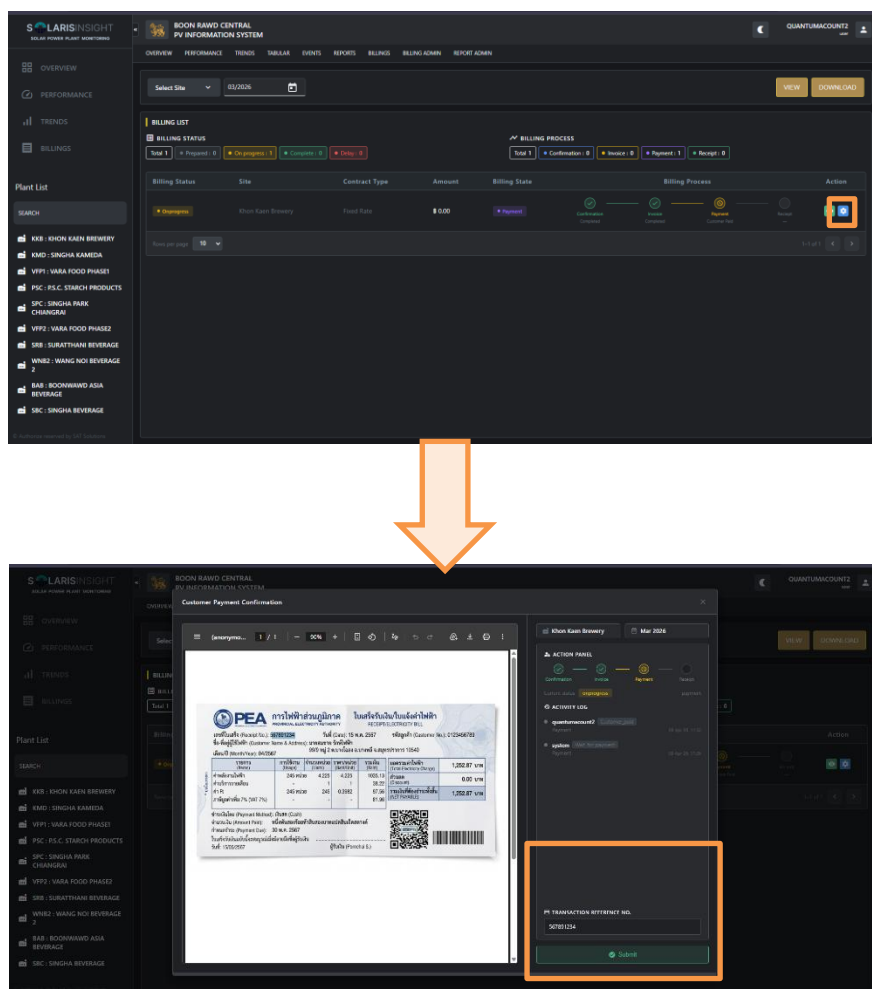
This step allows the internal team to verify the submitted payment details.

Process

1. Open the billing record via the **Action** button
2. Review the **payment proof document**
3. Verify:
 - Transferred amount matches the invoice
 - Transaction reference number
 - Payment details accuracy
4. Confirm the payment

Outcome

- Payment is verified and status updated to **Customer Paid**
- System proceeds to **Receipt Issuance**



4.8 Receipt Issuance (Step 4)

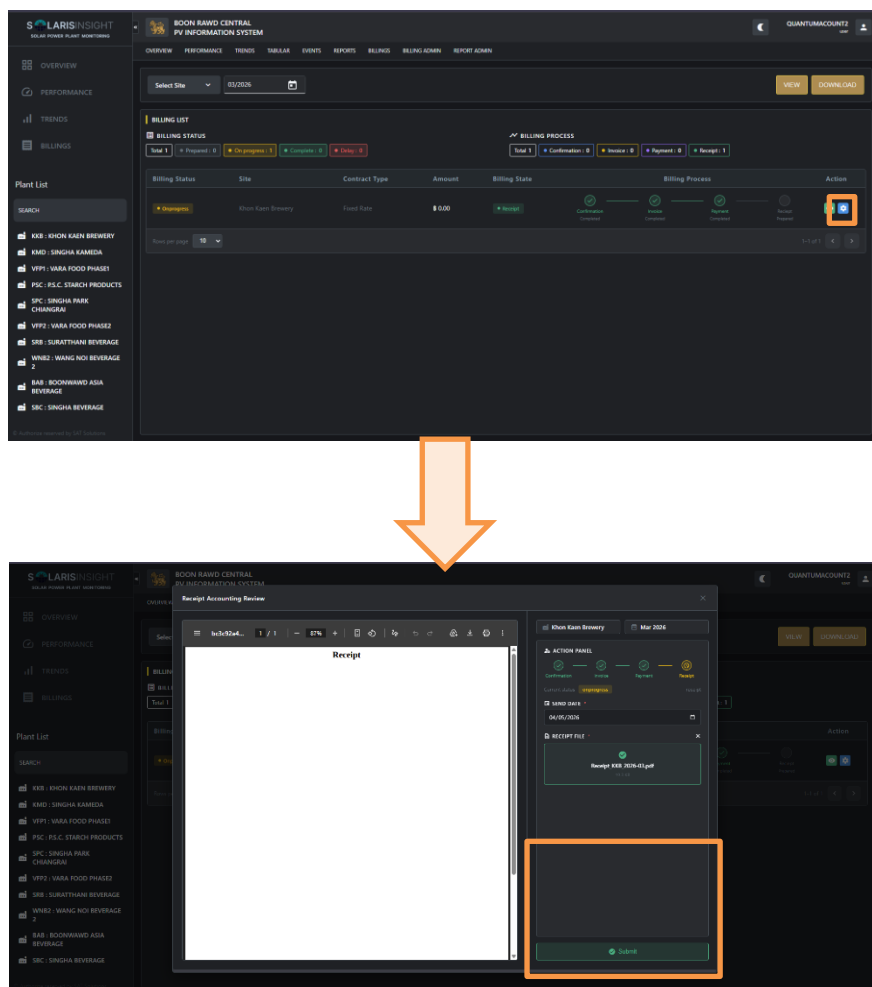
This step generates and finalizes the receipt after payment verification.

Process

1. Open the billing record via the **Action** button
2. Review the **Receipt document (PDF)**
3. Verify receipt details and payment reference
4. Submit to finalize the process

Outcome

- Receipt is issued to the customer
- Billing status updated to **Complete**
- Full billing cycle is completed





4.9 Final Billing Confirmation

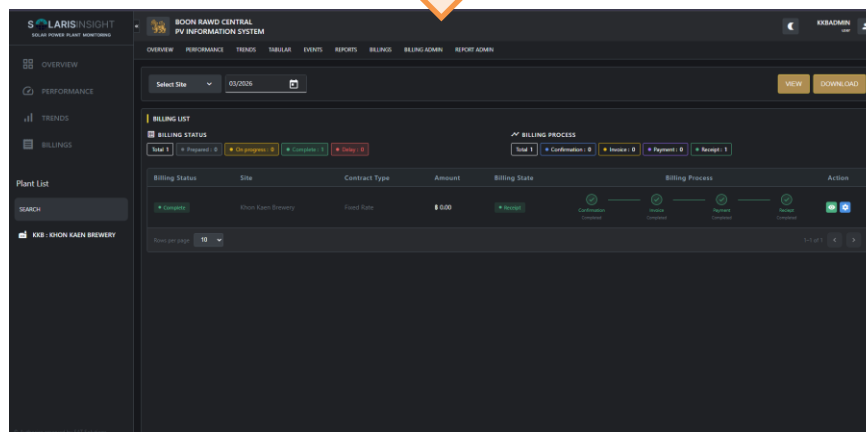
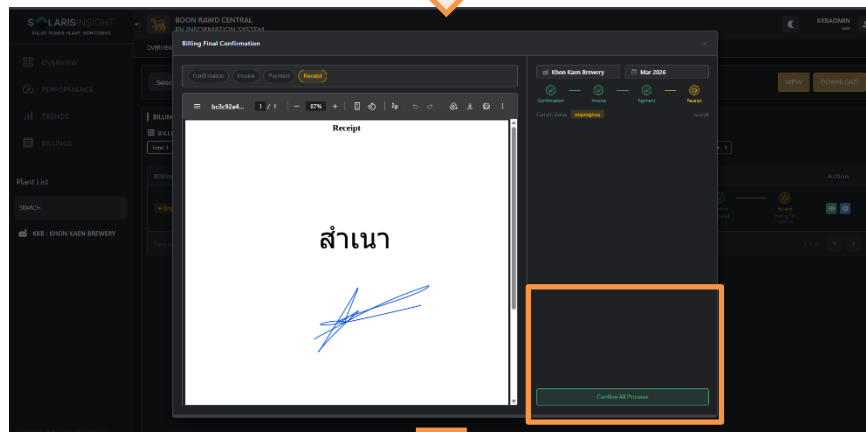
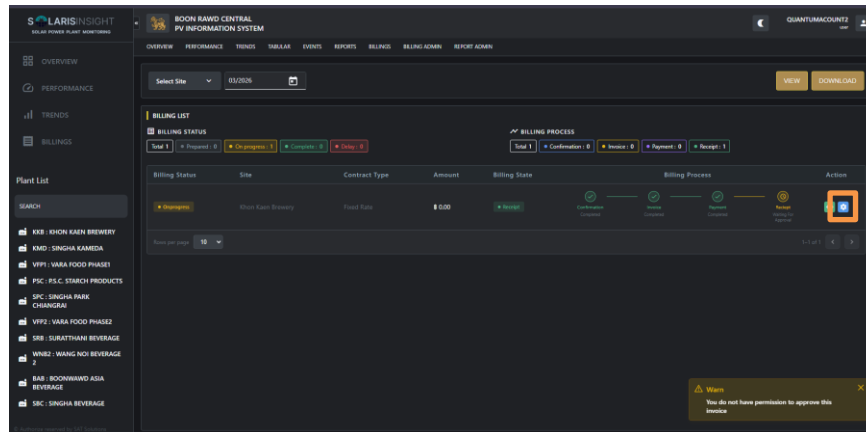
This step finalizes the entire billing process after all documents are completed.

Process

1. Open the billing record via the **Action button**
2. Review all completed documents (Confirmation, Invoice, Payment, Receipt)
3. Verify overall billing status and data consistency
4. Click **Confirm All Process**

Outcome

- Billing process is fully completed
- Status updated to **Complete**
- All documents are finalized and archived



4.10 Status & Monitoring

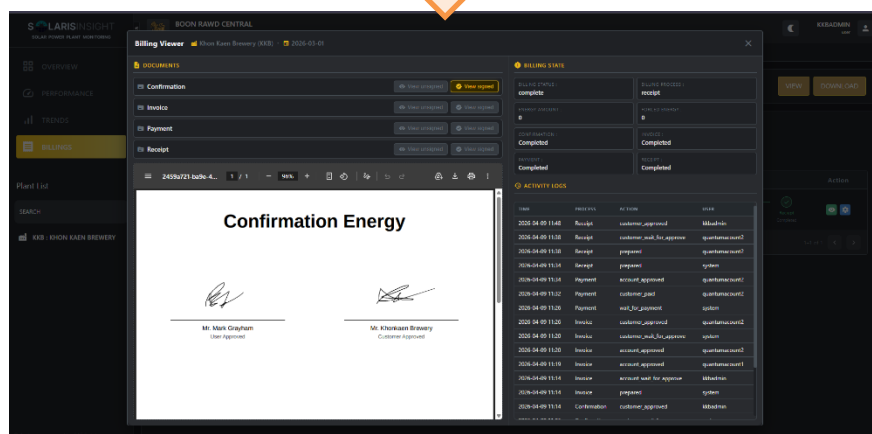
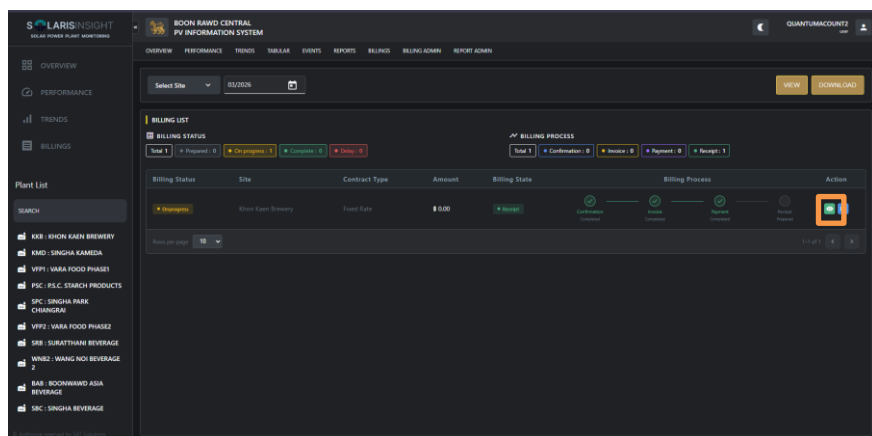
This module provides real-time tracking of billing progress and activity logs.

Features

- View current **billing status** and process stage
- Track progress across:
 - Confirmation
 - Invoice
 - Payment
 - Receipt
- Access all related documents in one place
- Monitor **activity logs** (user actions, timestamps, status changes)

Outcome

- Full visibility of billing lifecycle
- Improved transparency and traceability





4.11 Billing Management (Admin Control)

This module allows authorized users to manually update and manage billing data.

Features

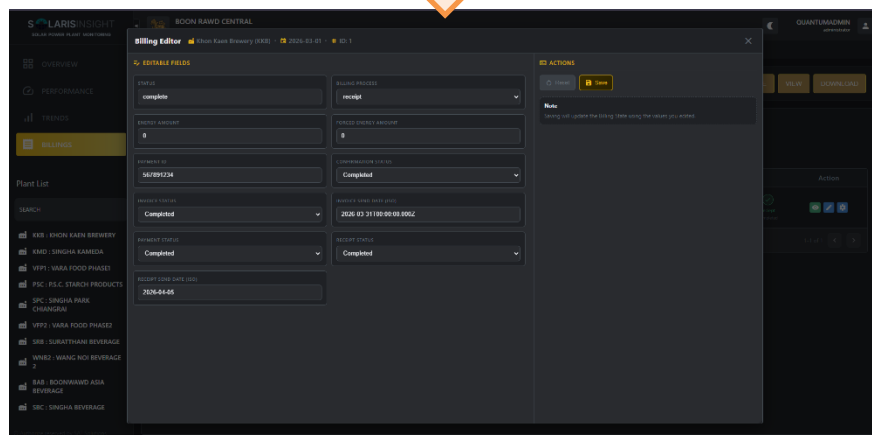
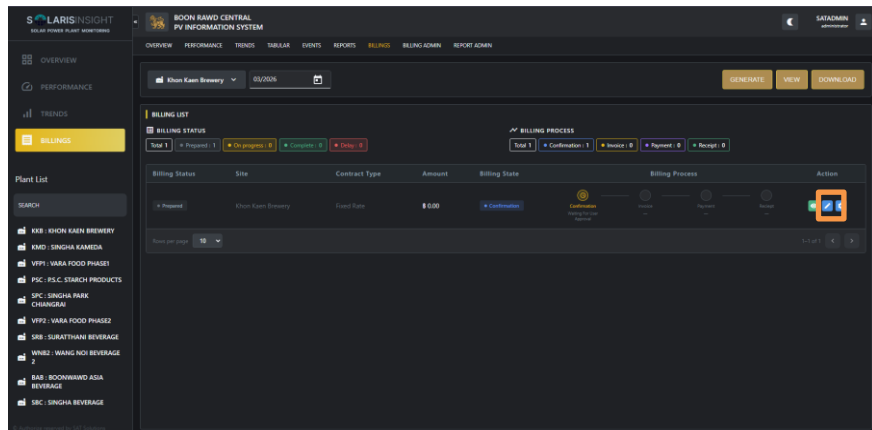
- Edit billing information (status, amount, process state)
- Update workflow stages manually if required
- Modify key fields such as:
 - Billing status
 - Billing process
 - Payment / receipt status
- Save changes with immediate system update

Use Cases

- Correct incorrect data
- Handle exceptional scenarios
- Override system workflow when necessary

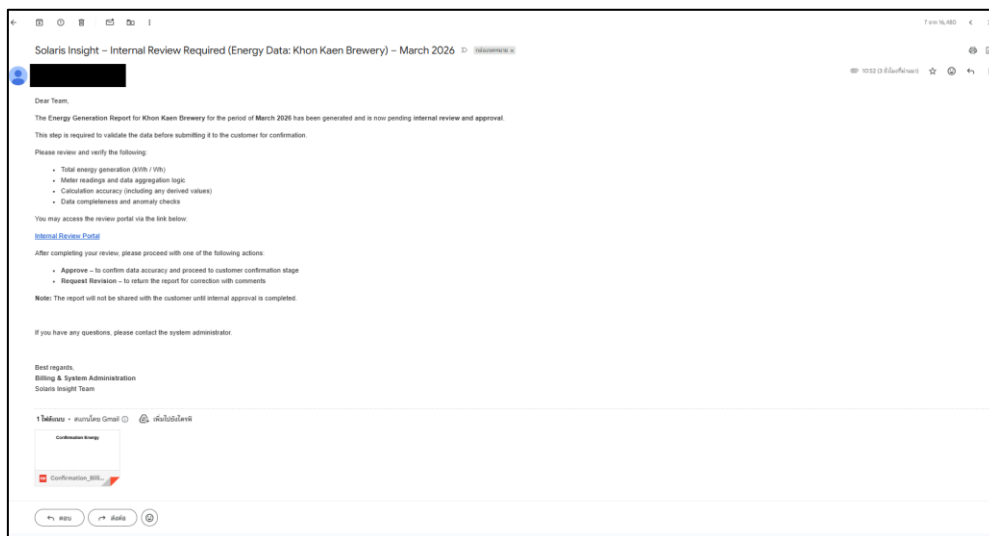
Outcome

- Billing data is updated in real-time
- Ensures flexibility while maintaining system control



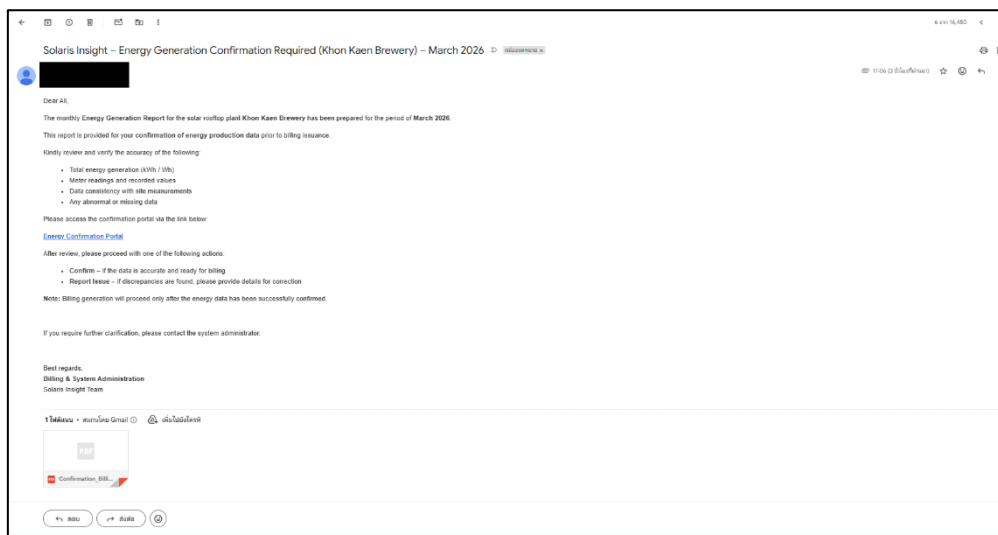
4.12 Notification & Email Communication

The system automatically sends email notifications at each stage.



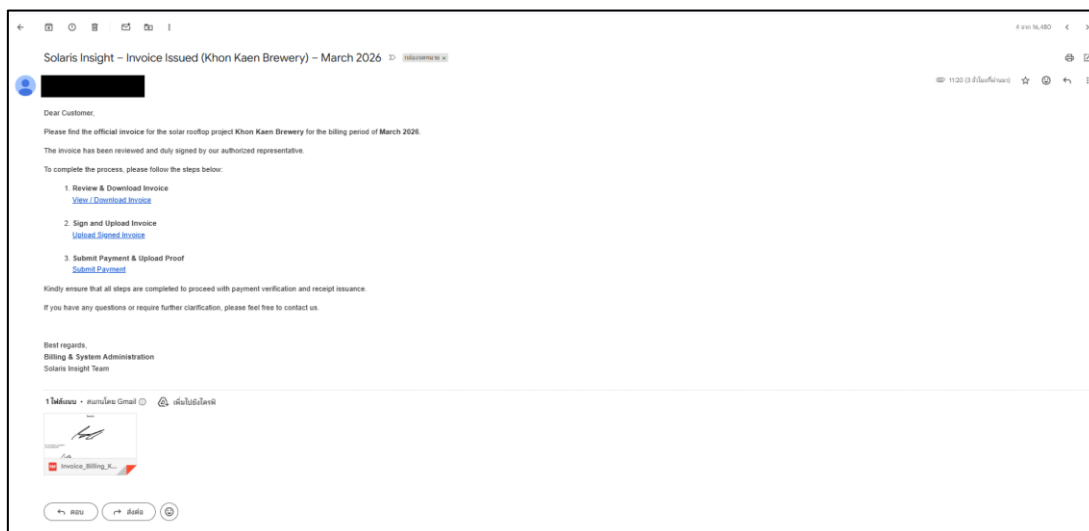
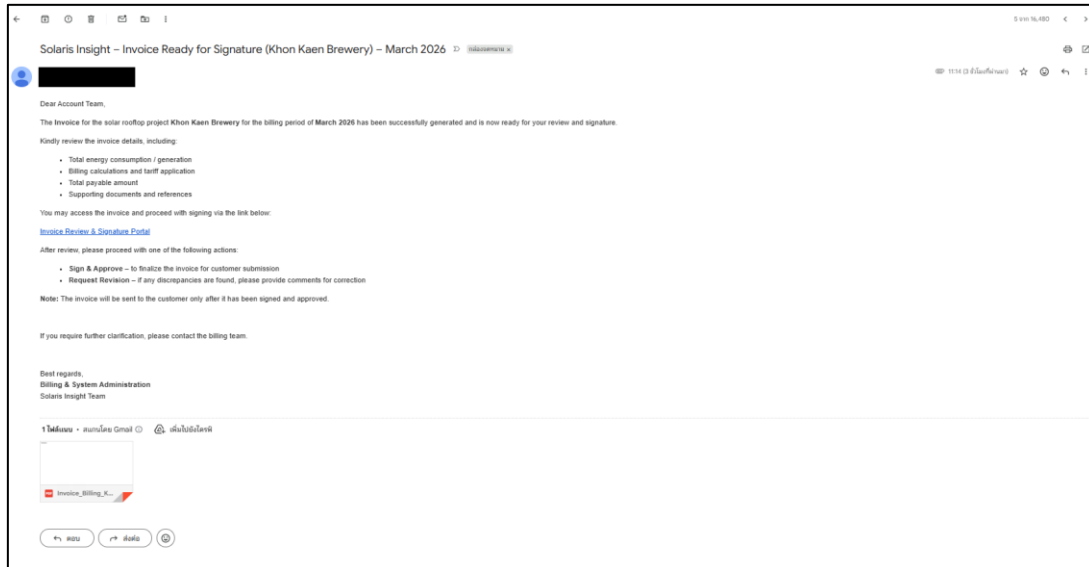
Internal Review Notification

Triggered when energy data is ready for internal validation.



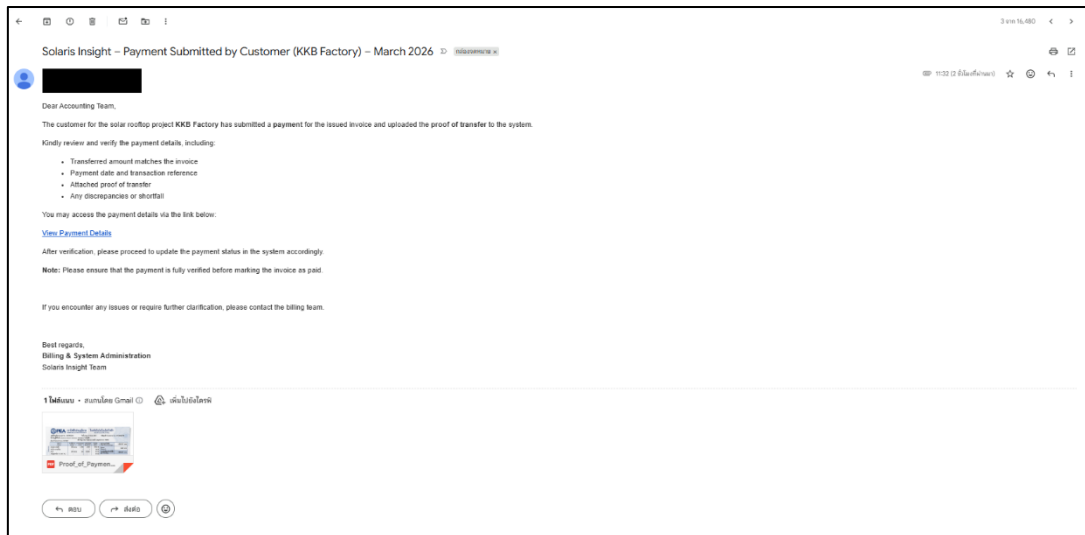
Customer Confirmation Request

Sent to customer for energy approval.



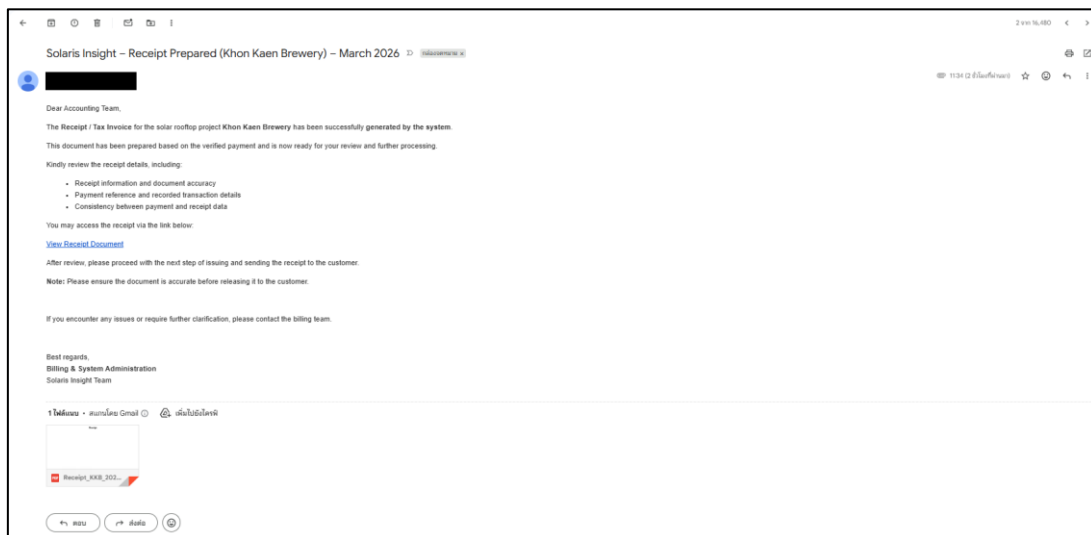
Invoice Issued Notification

Sent when invoice is finalized and ready for customer action.



Payment Submission Notification

Triggered when customer uploads payment proof.



Receipt Issued Notification

Final notification after billing completion.

4.13 Example Billing Design



หนังสือแจ้งค่าไฟฟ้าระบบโซลาร์เซลล์ ประจำเดือน มกราคม 2569

วันที่ 1 มกราคม 2569

เรื่อง แจ้งค่าไฟฟ้า
เรียน Sat solutions co. ltd

ตามรายละเอียดดังนี้

หมายเลขผู้ใช้ไฟฟ้า	รหัสเครื่องวัด	ประเภทอัตรา	แรงดัน	วันที่อ่านหน่วย
90730200003585484	250613870	4212	115kV	08 พฤศจิกายน 2568

พลังงานไฟฟ้า จากโซลาร์เซลล์ (หน่วย)	หน่วย	จำนวนเงิน (บาท)	ค่าอัตราต่อหน่วย (บาท)	3.80
	687448	74,037.98	ค่าอัตราต่อหน่วยลด 46.1% (บาท)	1.75
		74,037.98	รวมเป็นเงิน(บาท)	74,037.98
			VAT 7%	5,182.66
			รวมเงินที่ต้องชำระ(บาท)	79,220.64

รวมเงินที่ต้องชำระ เจ็ดหมื่นเก้าพันสองร้อยยี่สิบบาทหกสิบสี่สตางค์
โปรดชำระภายใน 30(สามสิบ) วันนับตั้งแต่ได้รับใบแจ้งหนี้ค่าพลังงานไฟฟ้า

หมายเหตุ
จึงเรียนมาเพื่อโปรดชำระเงินภายในวันกำหนดต่อไปด้วย จะขอบคุณยิ่ง

ขอแสดงความนับถือ

(ลงชื่อ).....ผู้ใช้ไฟฟ้าหรือตัวแทน
(.....)
ตำแหน่ง.....

ลงชื่อ.....
(.....)
ตำแหน่ง.....



Monthly Energy Report
December 2025
SINGHA KAMEDA SOLAR ROOFTOP



Energy Production:	824200.00 kWh	Energy Charge:	2431390.00 THB
Energy Production (On Peak):	588088.00 kWh	Energy Charge (On Peak):	2460501.38 THB
Energy Production (Off Peak):	236112.00 kWh	Energy Charge (Off Peak):	614764.81 THB
CO2 Reduction:	192153.99 Kg	Financial Saving (Discount):	0.00 THB
Tree Saving:	15907.06 Trees	Financial Saving (FT):	327372.24 THB

Day	Energy Production (kWh)	Energy On Peak (kWh)	Energy Off Peak (kWh)	Irradiance (Wh/m2)	Ambient Temp. (°C)	Module Temp. (°C)
01	32048.00	28496.00	3552.00	16230.34	24.70	29.46
02	31552.00	28104.00	3448.00	16062.00	25.65	30.71
03	31600.00	28304.00	3296.00	16051.21	27.49	31.96
04	26200.00	22600.00	3600.00	13719.63	27.78	30.58
05	18200.00	15752.00	2448.00	9018.84	27.52	28.93
06	31096.00	0.00	31096.00	15707.98	27.26	30.34
07	31256.00	0.00	31256.00	15903.96	26.39	29.60
08	32192.00	28992.00	3200.00	16308.60	25.67	27.71
09	27352.00	25144.00	2208.00	14028.88	25.23	27.14
10	32752.00	29448.00	3304.00	16616.52	25.88	29.72
11	7752.00	7200.00	552.00	3515.20	22.90	22.38
12	16544.00	16000.00	544.00	7979.01	24.37	26.16
13	19352.00	0.00	19352.00	9536.10	26.17	28.49
14	24856.00	0.00	24856.00	12596.51	27.01	29.13
15	31896.00	29496.00	2400.00	15974.19	26.81	30.72
16	31896.00	28696.00	3200.00	16211.22	26.47	30.37
17	30656.00	27600.00	3056.00	15658.98	26.28	29.24
18	31800.00	28600.00	3200.00	15671.41	26.90	29.18
19	32800.00	29752.00	3048.00	16484.07	25.97	29.41
20	33048.00	0.00	33048.00	16876.81	25.42	29.16
21	32552.00	0.00	32552.00	16647.21	25.41	28.10
22	29696.00	26600.00	3096.00	15675.60	26.80	29.79
23	22952.00	19952.00	3000.00	11791.83	27.84	30.21
24	28496.00	25648.00	2848.00	14789.27	27.62	31.16
25	29856.00	27152.00	2704.00	15159.61	28.41	30.62
26	33048.00	30152.00	2896.00	16589.87	25.98	28.44
27	800.00	0.00	800.00	537.58	23.23	20.68
28	48.00	0.00	48.00	10.17	25.35	20.79
29	31600.00	29000.00	2600.00	15171.48	26.30	29.92
30	30904.00	28400.00	2504.00	15239.79	25.47	29.70
31	29400.00	27000.00	2400.00	14153.29	25.47	28.80
Total/Average	824200.00	588088.00	236112.00	415917.14	26.12	28.66

